MSSCFormer: Multigranularity Spatial–Spectral Convolution Transformer Network for Hyperspectral Image Classification

Gang Chen, Lina Yang, *Member, IEEE*, Thomas Wu, *Senior Member, IEEE*, Lianhui Liang, *Member, IEEE*, Xichun Li, and Jun Li, *Fellow, IEEE*

***Abstract*— Recently, convolutional neural networks (CNNs) and Transformers have achieved considerable success in hyperspectral image (HSI) classification tasks. However, existing methods not only lack the study of spectral variability of samples from the same land class but also struggle to mine local–global spectral information and spatial structure information of HSI at different granularities effectively. To mitigate these limitations, this article proposes a multigranularity spatial–spectral convolution Transformer (MSSCFormer) network, which can reduce the intra-class spectral differences of samples and extract multigranularity spatial–spectral features from a local-global-local perspective. Specifically, MSSCFormer consists of three components: intra-class spectral attention (ICSA), spatial–spectral feature extractor (SSFE), and global–local convolutional Transformer (GLCT). First, ICSA redistributes the spectral weights of samples of the same land class by establishing an attention mapping between spectral channels within the class to reduce the intra-class spectral differences. Second, SSFE extracts shallow spatial features and multigranularity spectral features of HSI with reassigned spectral weights samples from a local perspective. Finally, GLCT takes advantage of CNNs and Transformers; it uses MHSA to model global spectral features and utilizes local context feature block (LCFB) to capture local spatial features in a multigranularity way. Experimental results on three benchmark datasets show that MSSCFormer exhibits excellent performance on the HSI datasets and outperforms state-of-the-art HSI classification algorithms.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) classification, intra-class spectral variability, multigranularity, transformer.**

Received 30 May 2025; revised 24 June 2025; accepted 11 July 2025. Date of publication 15 July 2025; date of current version 23 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62371144 and Grant 62461004 and in part by Guangxi Science and Technology Major Program under Grant AA23073019. *(Corresponding authors: Lina Yang; Thomas Wu.)*

Gang Chen is with the School of Computer, Electronics and Information, Guangxi University, Nanning, Guangxi 530004, China (e-mail: 2313394002@st.gxu.edu.cn).

Lina Yang is with the Institute of Scientific Research, Guangxi University, Nanning, Guangxi 530004, China (e-mail: lnyang@gxu.edu.cn).

Thomas Wu and Lianhui Liang are with the School of Electrical Engineering, Guangxi University, Nanning, Guangxi 530004, China (e-mail: xwu@gxu.edu.cn; Lianglh0308@126.com).

Xichun Li is with the School of Mathematics and Computer Science, Guangxi Normal University for Nationalities, Chongzuo, Guangxi 532200, China (e-mail: lixichun@gxnun.edu.cn).

Jun Li is with the Faculty of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: lijuncug@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3589190

I. Introduction

HYPERSPECTRAL imaging technology is an advanced tool capable of acquiring fine spectral information of a feature, which provides an important support for the study of the physicochemical properties of the Earth's surface through the precise acquisition of continuous narrow-band spectra [1], [2]. Hyperspectral image (HSI) consists of dozens to hundreds of spectral bands, with each pixel representing the reflectance or emission characteristics of the feature across the entire spectral range, thus creating a "spectral fingerprint" [3]. This uniqueness has led to a wide range of applications of hyperspectral technology in areas such as agricultural monitoring [4], target detection [5], environmental monitoring [6], and disaster management [7]. HSI classification aims to assign each pixel to a specific land-cover class. The rich spectral band information of HSI contributes to the accuracy of the classifications, but the high-dimensional spectral information may also lead to "dimensional catastrophe" and the Hughes effect [8], [9]. The wide range of applications of the HSI classification task and its complexity make it a hot research topic at present.

Considering the high-dimensional properties of HSI brings problems such as computational complexity and redundant features. Researchers have proposed a variety of methods to reduce spectral dimensionality and help extract discriminative low-dimensional feature representations [10], [11]. Overall, these methods can be classified into two classes: linear transformation-based dimensionality reduction methods, such as principal component analysis (PCA) [12], independent component analysis (ICA) [13], minimum noise fraction (MNF) [10], and linear discriminant analysis (LDA) [14]. However, the above methods are difficult to capture the nonlinear structure in HS, and to address this problem, researchers have proposed dimensionality reduction methods for nonlinear transformations, such as kernel principal component analysis (KPCA) [15], local linear embedding (LLE) [16], and manifold hypergraph learning (MHL) [17].

The exploration of various dimensionality reduction methods for HSI reveals that their fundamental goal is to enhance the efficiency and effectiveness of feature extraction. Traditional HSI classification algorithms rely on experts to

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

extract features through hand-designed methods, which are time-consuming and labor-intensive in complex scenes [18]. In contrast, machine learning methods have to some extent improved the inadequacy of hand-designed methods for feature extraction with typical classifiers (e.g., support vector machine (SVMs) [19], random forests (RFs) [20], nearest-neighbor classifiers [21]), but limited by the feature extraction capability of the models, these methods are still considered as shallow models [22].

In recent years, HSI classification algorithms based on deep learning have received extensive attention. Chen et al. [23] proposed stacked autoencoders (SAEs) to learn deep features and combine spatial–spectral information by stacking multiple AE layers. In addition, Chen et al. [24] proposed deep belief networks (DBNs) to fuse spatial and spectral information provided an important idea for later advanced learning methods. Recognizing the sequential nature of HSI, Mou et al. [25] analyzed the pixels of HSI as sequential data and applied recurrent neural networks (RNNs) for the first time to HSI classification. Extending the exploration of generative models, Chen et al. [26] combined generative modeling and classification tasks based on the generative ability of the diffusion model (Diff) to capture complex spectral distributions of HSI. Zhou et al. [27] proposed a two-branch long short-term memory network (LSTMs) to sequentially model the spatial and spectral information of HSI and fuse the features of both. Addressing the challenge of limited training data, Zhu et al. [28] initially introduced generative adversarial networks (GANs) to HSI classification tasks, employing the generator to synthesize HSI samples as a means of mitigating the challenge posed by limited training data availability. In parallel, Ding et al. [29] proposed self-correlated learning smoothly enhanced locality preserving graph convolution embedding clustering (S^2LGCC) that combines local preservation and autocorrelation mechanisms to enhance structure-aware representation without the need for labeling. Also, to better exploit spectral–spatial structural relationships, recent graph-based methods incorporate adaptive topology learning and self-supervised contrastive clustering to achieve effective classification even with limited annotations [30], [31], [32]. Meanwhile, Hu et al. [33] proposed a method based on 1-D CNN, which pioneered the use of CNNs for spectral feature modeling. To fully utilize the spectral and spatial information of HSI, Zhao and Hu [34] proposed a balanced local discriminant embedding algorithm to extract spectral features from high-dimensional HSI and use 2-D CNN to extract spatial features at a high level. Considering the 3-D nature of HSI, the advantages of 3-D convolutional kernel joint feature extraction are applied to HSI classification. For instance, He et al. [35] proposed a multiscale 3-D CNN framework, which enables the model to learn both 2-D multiscale spatial and 1-D spectral features in an end-to-end approach. Most recently, Zhang et al. [36] proposed PFS3F, a framework that probabilistically fuses superpixel-wise structural cues and semantic-aware features, achieving strong spatial consistency and refined class boundaries.

Although the above methods have achieved significant improvement in HSI classification tasks, they still have certain limitations, which are summarized as follows.

1) Generative architectures, such as SAEs and DBNs, are constrained by the physical scarcity and multidimensional complexity of HSI, and their hierarchical pre-training mechanisms and generative adversarial processes, such as GANs and Diff, suffer from the inherent problem of unstable training dynamics, which results in poor model convergence and efficiency.
2) RNNs and LSTMs face the problem of dimensional mismatch between sequence and spatial topology when modeling spectral dimensions. Its unidirectional recursive mechanism limits its spatial context modeling capabilities and makes it difficult to break through the long-range dependency learning bottleneck caused by gradient diffusion.
3) Although the local receptive field of CNN-based methods can capture spatial features, its translational invariance assumption fundamentally conflicts with the heterogeneous distribution of HSI. The fixed kernel structure is unable to adapt to the global contextual associations in spectral dimensions, and the semantic disconnections are easily generated when fusing multigranularity features, resulting in the degradation of inter-class separability.

Recently, the application of Transformer networks [37] in the field of computer vision (CV) has become a popular research topic. The application of vision Transformer (ViT) [38] in HSI classification is gradually maturing. It captures the long-distance dependence between different elements through the multihead self-attention mechanism (MHSA) to extract global features between spatial and spectral dimensions. Specifically, MHSA extracts important feature information through various matrix multiplication and addition operations. Models relying solely on CNN or Transformer structures are still difficult to achieve ideal classification results. Hybrid frameworks obtained by researchers fusing CNN and Transformer have further advanced the field of HSI classification. Most of the existing hybrid CNN-Transformer methods use two branches to extract local and global information of the data, respectively. For example, Yang et al. [39], Zhang et al. [40], and Zhou et al. [41] fused CNN and Transformer networks to achieve simultaneous extraction of local spatial features and long-range dependencies in images. However, these methods also have some limitations.

1) Although some of the improved Transformers have attempted to capture both spectral and spatial information, the fusion of spectral and spatial features is often not tight enough in these improved Transformers, making it difficult for the models to effectively capture the complex relationship between the two.

2) CNN-Transformer-based methods do not make full use of the CNN for spatial feature extraction. And the Transformer part lacks the ability to mine localized information, which leads to the inability to adequately obtain global and local information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

3) The difference in spectral information of the same land class samples will have some influence on the final classification results. However, the existing methods lack research on the spectral difference of the same class samples, which can lead to a limited ability of the model to extract more discriminative spectral features.

4) Existing methods exhibit insufficient extraction of spectral information and lack the ability to extract shallow spectral information before modeling the spectral information using a Transformer, leading to inadequate mining of spectral information by the model.

Aiming at the above-mentioned problems, we propose a multigranularity spatial–spectral convolution Transformer network (MSSCFormer). To reduce the differences between the spectral information of samples of the same class, an intra-class spectral attention (ICSA) module is designed. Furthermore, a spatial–spectral feature extractor (SSFE) consisting of a 3-D depth-separable convolution with residual concatenation (3-D DSConv) and a multigranularity spectral feature fusion module (MGSFF) is designed to extract more discriminative spatial–spectral features. Finally, a global local convolution Transformer (GLCT) is designed, which extracts the global information of HSI using MHSA and extracts the local information at multigranularity using local context feature block (LCFB). The main contributions of this article are as follows.

1) A local-global-local model, MSSCFormer, is proposed, which fully exploits the advantages of CNN and Transformer by fusing spatial–spectral features, global–local features, and multigranularity features. SSFE extracts short-range band dependencies in adjacent spectral bands, and GLCT models the long-range band interactions, which meet the need for spectral continuity. The sensitivity of the model to edges and textures is enhanced by a secondary local optimization of the LCFB to accommodate the heterogeneity of the spatial distribution.
2) A novel module, ICSA, is designed, which redistributes spectral weights to samples of the same land class through the pooling and convolution operations of two branches and combines with learnable class-centered constraints to reduce intra-class spectral differences, so as to enable MSSCFormer to better distinguish the features of different classes of samples. ICSA breaks through the limitations of the traditional attention mechanism, which only focuses on the enhancement of discriminative features, and realizes the adaptive correction of the spectral distribution of the same class samples.
3) An innovative SSFE is proposed, a cascaded spatial–spectral feature extraction architecture from a local perspective. It extracts the shallow spatial and spectral features first. Then, by a four-way heterogeneous convolutional kernel structure, the short-range neighborhood correlation and long-range trend features are captured in parallel, and the residual cross-layer connections form a spectral detail enhancement pathway, which significantly improves the model's ability to identify weak spectral features.
4) A novel GLCT is designed from a global–local perspective, which integrates the global attention modeling of Transformer with the multigranularity local feature extraction of CNN in a temporal order. We use the central pixel as the classification token to establish global dependency, and enhance the extraction of edge texture features through the heterogeneous convolutional combination of LCFB module, which ultimately forms an iterative optimization mechanism of "global guides the local, and the local corrects the global."

The rest of the article is organized as follows. Section II reviews related work. Section III describes the structure of the model and each functional module in detail. Section IV describes the datasets used in the model, the relevant configurations for the experiments, and the experimental results. Section V summarizes our overall work.

II. Related Work

In this section, we give a brief summary of the main approaches in the HSI classification work. This section focuses on the following three approaches: methods based entirely on CNNs for HSI classification, methods based entirely on Transformers for HSI classification, and methods based on a hybrid CNN-Transformer for HSI classification.

A. Methods Based on CNNs for HSI Classification

Due to the limitations of traditional manual-based feature extraction methods in specific application scenarios, researchers have started to explore deep learning-based HSI classification methods. The high efficiency of CNNs in spatial feature extraction and the excellent properties of convolutional operations (e.g., local awareness, translation invariance, and weight sharing) have led to a wide range of applications of CNN-based methods in the field of HSI classification. HSI classification algorithms for CNNs are broadly classified into three types: 1-D CNN-based, 2-D CNN-based, and 3-D CNN-based. Hu et al. [33] proposed a 1-D CNN-based HSI classification algorithm, which pioneered the use of CNNs for HSI classification, and the method consists of convolutional, pooling, and fully connected layers. It uses a five-layer 1-D CNN to capture the spectral information of the image, but lacks the ability of spatial feature extraction. Zhao and Hu [34] proposed a 2-D CNN to make up for the lack of spatial feature extraction ability of the 1-D CNN. However, the 2-D CNN is limited by the local receptive field of the convolution kernel and is still not able to extract spatial–spectral features efficiently. He et al. [35] proposed a multiscale 3-D CNN framework, which enables the model to jointly capture both spectral and spatial features. Zhong et al. [42] proposed a spectral–spatial residual network (SSRN), which utilizes the idea of residual learning to obtain richer spatial–spectral features by using the residual connection in conjunction with 3-D convolutional operations. Thereafter, Li et al. [43] designed a dual-branch dual-attention mechanism network (DBDA), which captures features by incorporating spatial and channel attention mechanisms into the spatial and spectral feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

branches, respectively, and fusing the features obtained from the two branches. Meanwhile, Zhu et al. [44] proposed a residual spatial–spectral attention network (RSSAN), which strengthens the extraction of key information by incorporating both the attention mechanism and residual learning into the extraction of spatial–spectral features. Zhang et al. [45] proposed a dual-branch CNN framework that separately extracts global spectral features using pyramid texture filters and local spatial structures via superpixel-based analysis, effectively enhancing structural representation in HSI classification. The above-mentioned related methods have achieved good results in HSI classification tasks; however, the inherent limitation of convolutional kernels in capturing global contextual information imposes significant challenges for this class of methods.

B. Methods Based on Transformer for HSI Classification

Transformer's excellent performance in the field of natural language processing makes it possible for potential development in the field of CV. Dosovitskiy et al. [38], who introduced the Transformer into the CV domain, have led researchers to explore the Transformer for further research in the field of conventional image processing and in the field of HSI classification. Unlike the local sense field of CNN, ViT models the global relationship between different elements in an image and thus obtains the global information of the image through MHSA. He et al. [46] proposed HSI-BERT, a bidirectional encoder structure that captures global information regardless of spatial distance by a global receptive field. Qing et al. [47] designed an end-to-end model, SATNet, which extracts the key information of the spectral bands of an image through the spectral attention mechanism, and captures the global features of the image by using the self-attention mechanism. Furthermore, Hong et al. [48] proposed a model purely using Transformer, Spectralformer, which learns the local sequence information through grouped spectral embeddings to produce grouped spectral embeddings. In addition, it uses a cross-layer adaptive fusion module to effectively convey valuable information across layers. Meanwhile, Xue et al. [49] introduced SPRLT-Net, which uses a local Transformer to dynamically obtain the spatial attention weights and get a continuous subpatch sequence by a spatial partition restoration. Xu et al. [50] proposed SS1DSwinTransformer, combining 1DSwin Transformer and group feature tokenization. It captures HSI's long-range dependencies via window attention and shift window operations, and converts features into tokenized representations for classification using group feature tokenization. The models that rely exclusively on Transformers have obvious advantages in modeling global features, but they lack the capacity to fully utilize the rich spatial information of HSI. Recent breakthroughs in state-space models have opened new frontiers for HSI sequence processing. The Mamba architecture [51] introduced an interval group S6 mechanism to partition spectral bands into multiple nonoverlapping groups at intervals and integrate a unidirectional S6 with a specific scanning direction for each group to achieve nonredundant sequence modeling. Building upon this, He et al. [52] proposed 3DSS-Mamba, which utilizes a 3D-spectral–spatial selective scanning (3DSS) mechanism to perform pixel-wise selective scanning on 3-D hyperspectral tokens across spectral and spatial dimensions and uses five scanning routes to explore the impact of different scanning directions. Though promising, these methods still face challenges in preserving spatial structural relationships during sequence unfolding operations.

C. Methods Based on Hybrid CNN-Transformer for HSI Classification

Models relying exclusively on CNN or Transformer have their own drawbacks on HSI classification tasks, and researchers have taken full advantage of CNN and Transformer by combining them to build novel hybrid frameworks. Zhang et al. [40] proposed a convolutional-Transformer mixer (CTMixer), which captured local spatial–spectral information with a set of parallel residual blocks. The obtained features were passed through a parallel two-branch structure consisting of a CNN and a Transformer to extract local–global features of the HSI. Similarly, Yang et al. [53] introduced an HSI Transformer (HiT), which used spectral-adaptive 3-D convolution and a convolution permutator to produce spatial–spectral information. Sun et al. [54] proposed a spatial–spectral tokenized Transformer (SSFTT) that extracts shallow spatial–spectral features of HSI by using convolution in 2-D and 3-D. Then, the transformed features by the Gaussian-weighted feature tokenizer were input into the transformer encoder for global information learning. Furthermore, a learnable spectral and spatial morphological network (MorphFormer) was introduced in [55], which innovatively introduced a morphological attention mechanism that used morphological operations of dilation and erosion to capture the local structure and spatial relationships of pixels. Liang et al. [56] proposed a two-branch structure that utilizes a CNN augmented with the SimAM attention mechanism that focuses on extracting local spatial features, and a Transformer branch that is responsible for capturing global contextual information and modeling long-range spectral dependencies. Meanwhile, Ouyang et al. [57] proposed a Hybrid Transformer model (HybridFormer), which first captured fine-granularity and coarse-granularity features. Then, a spatial–spectral attention mechanism is introduced to weight the spatial and spectral domains, respectively, and this mechanism can accurately select key features from HSI. In [58], a memory-augmented spectral–spatial transformer (MASSFormer) was investigated, which stored prior knowledge by employing the memory tokenizer to transform the features into memory tokens and incorporated the memory-augmented transformer encoder to achieve full information fusion while mitigating potential depth decay in models. In addition, Zhou et al. [41] proposed a dual-branch model (DCTN) with CNN for spatial features and Transformer for spectral features. An efficient interactive self-attention (SA) mechanism was designed in the Transformer branch to ensure information flow between spatial and spectral features. This interactive design allowed the model to learn how spatial information affects spectral information and vice versa.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

Although these works have achieved some results, there are still some problems. How to effectively distinguish spectral differences between samples of the same land class. How to make full use of CNN's local context mining ability and Transformer's long-range modeling ability. How to extract multigranularity spatial–spectral information. These problems limit the further improvement of model performance.

III. Proposed Method

In this section, we will describe the overall structure of the MSSCFormer method, which includes the details of the ICSA module, SSFE module, GLCT module, and LCFB module.

A. Overview of MSSCFormer

In this work, a new cascade local-global-local hybrid convolutional Transformer for HSI classification, named MSSCFormer, is proposed. As shown in Fig. 1, the proposed MSSCFormer consists of three modules: ICSA, spatial spectral feature extractor (SSFE), and GLCT. The ICSA module is first designed to reduce the spectral differences between samples of the same land class for the subsequent extraction of more representative features. The SSFE module consists of a 3-D depth-separable convolution with residual concatenation and a multigranularity spectral feature fusion module. It is mainly used to extract shallow spatial features and multigranularity spectral features. GLCT consists of MHSA and LCFB modules. MHSA is used to establish long-range dependencies between pixels to capture the global information of the image. LCFB consists of multigranularity 2-D convolution and residual concatenation to make up for the lack of the Transformer's ability to mine local information. Experimentally, it is proven that the three modules are effective in working together. Fig. 1 overviews the proposed MSSCFormer.

B. Intra-Class Spectral Attention

To reduce the spectral differences between samples of the same land class, we propose a novel module, ICSA, for HSI classification tasks. Different from the traditional spectral attention mechanism, this module can assign each spectral band with different weights through the attention mechanism. Moreover, we design a spectral feature alignment loss function to constrain the spectral weight change process for the purpose of reducing intra-class spectral differences. Also, ICSA provides a great basis for the subsequent extraction of more discriminative features.

The traditional spectral attention mechanism assigns different weights to the spectral bands of samples of the same class, and we found that this practice makes the intra-class disparity larger, which has an impact on the classification results. ICSA reduces the intra-class disparity of samples of the same class by reassigning the spectral weights of the samples. As shown at the top left of Fig. 1, in HSI classification tasks, the original image will be divided into multiple same-sized HSI patches. We denote the ith HSI patch by $\mathbf{P}_i$, where $\mathbf{P}_i \in \mathbb{R}^{s\times p\times p}$, s and p denote the number of spectral dimensions and spatial size of the HSI patch, respectively. To avoid the interference of spatial information with spectral information, we use adaptive maximum pooling (AMP) and adaptive average pooling (AAP) operations to generate two feature vectors with different spatial information. They are expressed as $\mathbf{F_{max}} \in \mathbb{R}^{s\times 1\times 1}$, $\mathbf{F_{avg}} \in \mathbb{R}^{s\times 1\times 1}$, i.e.,

$$\mathbf{F_{max}} = \mathrm{AMP}(\mathbf{P}_i) \quad \text{and} \quad \mathbf{D_{avg}} = \mathrm{AAP}(\mathbf{P}_i). \tag{1}$$

Specifically, adaptive maximum pooling and adaptive average pooling are used to extract the global spectral information and local spectral information of the $\mathbf{P}_i$. Adaptive maximum pooling can effectively capture the most significant features in the spectrum, which helps to preserve the detailed information of the image, while adaptive mean pooling filters some noise by averaging, thus effectively capturing the global information in the spectrum. An effective representation of the spectral information is obtained by fusing the two information. The two obtained feature representations are then fed into the channel attention module, an attention mechanism consisting of two 2-D convolutions with a convolution and a ReLU activation function with the aim of obtaining two different weight representations, $\mathbf{w}_{max}$ and $\mathbf{w_{avg}}$. Combine $\mathbf{w_{max}}$ and $\mathbf{w_{avg}}$ by element-wise addition to obtain the spectral weight $\mathbf{w_i}$, i.e.,

$$\begin{aligned}\mathbf{w}_i = \sigma(&\mathbf{W}_{\mathrm{conv2}} * \mathrm{ReLU}(\mathbf{W}_{\mathrm{conv1}} * \mathbf{F}_{\max}) \\ &+ \mathbf{W}_{\mathrm{conv2}} * \mathrm{ReLU}\big(\mathbf{W}_{\mathrm{conv1}} * \mathbf{F}_{\mathrm{avg}}\big)\big)\end{aligned} \tag{2}$$

where $\sigma(\cdot)$ represents the Sigmoid function to rescale the spectral weight between 0 and 1. ReLU$(\cdot)$ is a nonlinear activation function. $\mathbf{W}_{\mathrm{conv1}}$ and $\mathbf{W}_{\mathrm{conv2}}$ denote the weights of the two 2-D convolution layers. $\mathbf{W}_{\mathrm{conv1}} \in \mathbb{R}^{(s/\beta)\times s}$, (s/β) is the output dimension, s is the input dimension. $\mathbf{W}_{\mathrm{conv2}} \in \mathbb{R}^{s\times(s/\beta)}$, s is the output dimension, and (s/β) is the input dimension. β is a reduction ratio used to reduce the computational overhead. In this method, we set β to 8. * represents the above 2-D convolution layers with kernels equal to 1.

Finally, after the various operations described above, we can obtain the spectral weights $\mathbf{w_i}$ corresponding to the input image patch $\mathbf{P_i}$, which indicates the importance of different spectral bands in $\mathbf{P_i}$. For this block, if no constraints are set may cause the model to learn some features that are not related to or distinguishable from the class, and it is difficult to focus on the bands that are specific to the class. Therefore, to better reduce the intra-class differences between samples of the same land class, we introduce the spectral feature alignment $\mathcal{L}_s$ loss to constrain the spectral weight change process of ICSA. Specifically, it calculates the loss by calculating the squared Euclidean distance between the embedding vector of each sample and the embedding vector of its corresponding spectral weight center. The formula for calculating the squared distance between the spectral weights and the class center for a single sample is as follows:

$$\mathcal{D}_{ik} = ||\mathbf{x}_i - \mathbf{c}_k||^2 = ||\mathbf{x}_i||^2 + ||\mathbf{c}_k||^2 - 2\mathbf{x}_i{}^T\mathbf{c}_k \tag{3}$$

where $\mathbf{x}_i$ represents the input sample i, $\mathbf{c}_k \in \mathbb{R}^d$, and k and d denote the learnable central vector of class k and the embedding vector, respectively. $||\cdot||$ represents the Euclidean norm.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

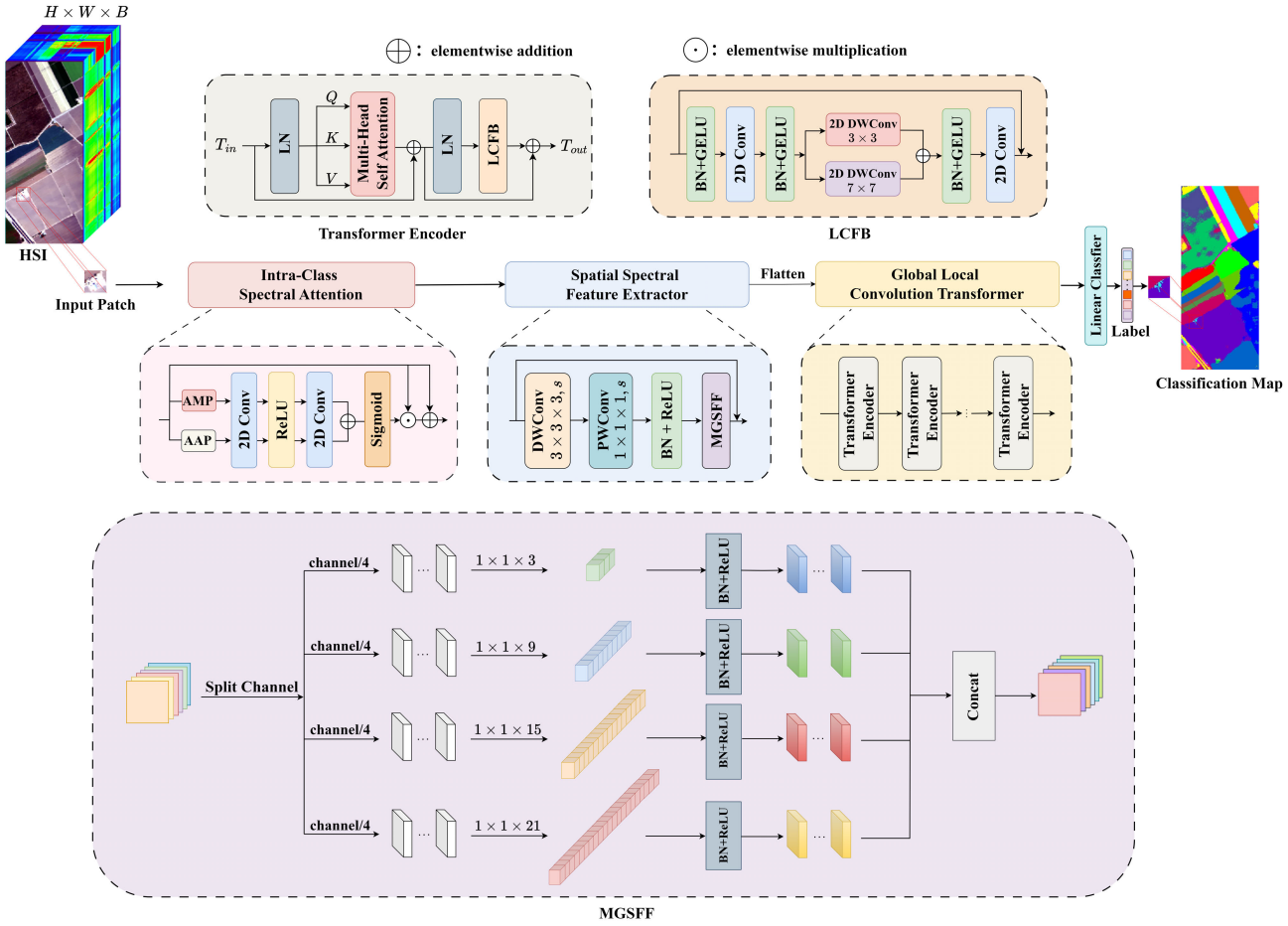


Fig. 1. Overview of the proposed MSSCFormer.

The calculation for the entire batch of data is as follows:

$$\mathcal{D} = \begin{bmatrix} ||\mathbf{x}_1||^2 & \cdots & ||\mathbf{x}_1||^2 \\ ||\mathbf{x}_2||^2 & \cdots & ||\mathbf{x}_2||^2 \\ \vdots & \ddots & \vdots \\ ||\mathbf{x}_B||^2 & \cdots & ||\mathbf{x}_B||^2 \end{bmatrix} + \begin{bmatrix} ||\mathbf{c}_1||^2 & \cdots & ||\mathbf{c}_M||^2 \\ ||\mathbf{c}_1||^2 & \cdots & ||\mathbf{c}_M||^2 \\ \vdots & \ddots & \vdots \\ ||\mathbf{c}_1||^2 & \cdots & ||\mathbf{c}_M||^2 \end{bmatrix} - 2\mathbf{X}\mathbf{C}^T \tag{4}$$

where the entire batch of data $\mathbf{X} \in \mathbb{R}^{B\times d}$, B represents the batch size, central matrix $\mathbf{C} \in \mathbb{R}^{M\times d}$, and M represents the number of land classes.

The loss function is expressed as follows:

$$\mathcal{L}_s = \frac{1}{B}\sum_{i=1}^{B}\left|\left|\mathbf{w}_i - \mathbf{c}_{\mathbf{y}_i}\right|\right|^2 \tag{5}$$

where y_i represents the true land-cover class of sample i, $1 \le y_i \le M$, and M denotes the number of classes. $\mathbf{c}_{y_i}$ denotes the center of spectral weight of the y_ith class, which is a learnable parameter in experiments. $\mathbf{c}_{y_i}$ is randomly sample through a Gaussian distribution, with each class corresponding to a center vector. During training, it is automatically updated by backpropagation and optimizer, with the goal of bringing samples of the same class closer to their respective class centers, improving the clustering and classification of features. From a mathematical point of view, this loss function introduces the idea of "centralization." By minimizing $\mathcal{L}_s$, the spectral weights of samples of the same class will gradually approach the center of their class, forming a tight distribution. This "centralization" effect reduces the dispersion of samples of the same class and makes the distribution of different classes easier to distinguish. Simply put, it reduces the intra-class differences and increases the inter-class differences. We multiply the updated weights by the weights of the original HSI element by element, and then add the result element by element to the original weight to obtain the final weight $\mathbf{P}_i'$ as follows:

$$\mathbf{P}_i' = \mathbf{P}_i \odot \mathbf{w}_i + \mathbf{P}_i \tag{6}$$

where $\odot$ represents element-by-element multiplication.

C. Spatial Spectral Feature Extractor

After narrowing the differences between the spectral bands of the same land class samples, we can better mine the spatial information and spectral information of HSI. We design a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

module named SSFE. The spatial dimension of HSI contains local features such as texture and shape, while the data in the spectral dimension shows long-range dependence and diversity. Based on these 2-D thinking, we design two modules in SSFE to capture shallow spatial features and multigranularity spectral features from a local perspective.

The overview diagram of the SSFE module is shown in the middle part of Fig. 1. It consists of two main parts: the DSConv and the multigranularity spectral feature fusion (MGSFF). We first feed the $\mathbf{P}_i'$ obtained after ICSA to the DSConv module to extract shallow spatial features and multigranularity spectral features. However, its expressive power in the spectral dimension is limited, especially in cross-granularity feature capture. A single-granularity feature extraction approach may not be sufficient to fully reveal the inherent richness and multilevel nature of spectral data. Therefore, we immediately adopt MGSFF, a module that focuses on spectral dimension feature extraction, to accurately capture the spectral information of $\mathbf{P}_i$ at multiple granularities, which effectively compensates for the limitations of DSConv in spectral dimension feature extraction.

The structure of the DSConv is very simple, which adopts 3-D depth-separable convolution and residual connection techniques, as shown in the middle part of Fig. 1. By kernel size of 3 DW Conv and kernel size of 1 PW Conv, the model not only captures the spatial structure information and spectral information of the image, but also greatly reduces the parameters of the model and the computational complexity compared to the standard convolution with a kernel size of 3, which is conducive to achieving fast classification in practical applications. The computational complexity of the depth-separable convolution and the ordinary convolution is as follows:

$$\mathbf{C}_{\mathrm{DW}} = \mathbf{C}_{\mathrm{in}} * (K * K * K) * D * H * W \tag{7}$$

$$\mathbf{C}_{\mathrm{PW}} = \mathbf{C}_{\mathrm{in}} * \mathbf{C}_{\mathrm{out}} * D * H * W \tag{8}$$

$$\mathbf{C}_{\mathrm{DSConv}} = \mathbf{C}_{\mathrm{DW}} + \mathbf{C}_{\mathrm{PW}} \tag{9}$$

$$\mathbf{C}_{\mathrm{ordinary}} = \mathbf{C}_{\mathrm{out}} * \mathbf{C}_{\mathrm{out}} * (K * K * K) * D * H * W \tag{10}$$

where $\mathbf{C}_{\mathrm{in}}$ and $\mathbf{C}_{\mathrm{out}}$ represent the number of input channels and output channels, respectively. K represents the size of the convolution kernel. D represents the number of spectral bands. H and W represent the height and width of the input image, respectively. In addition, the residual linkage technique is used to fuse the original inputs of DSConv and the results after feature extraction to alleviate the model gradient vanishing as well as to enhance the feature learning capability of the model.

Although the DSConv module can extract information in both spatial and spectral dimensions of an image, it mainly focuses on local features and is unable to extract rich spectral features effectively. For the exploration of spectral features, we design the MGSFF module, as shown in the bottom part of Fig. 1, which achieves deep mining and extraction of spectral dimensional features through multigranularity convolution operation. HSI usually contain multiple bands, and each band contains different spectral information. MGSFF uses group convolution to divide the input channels into four groups equally, and each group processes a part of the spectral channels independently. It can reduce the inter-channel interference and improve the efficiency and accuracy of information extraction. Then, 3-D convolution with different convolution kernel sizes is used for each of the four groups. Small convolution kernels are used to capture short correlations between bands, that is, fine-grained information between neighboring bands in the spectral dimension, and these small convolution kernels are able to extract local inter-band features more finely, focusing on local spectral details. Large convolution kernels are used to capture long-range correlations, that is, correlations between spectral bands that are far apart. These large convolution kernels can better extract global information between different bands and help the model understand the changing trend of the entire spectral dimension. The MGSFF module can significantly improve the representation of spectral information by combining convolutional kernels of different sizes. Each convolutional layer is followed by batch normalization (BN) and ReLU activation function. The BN can effectively alleviate the problems of gradient vanishing and gradient explosion during training, which makes the network training more stable. The ReLU activation function can effectively activate the complex features in HSI, which helps to capture the nonlinear spectral features. Finally, the features extracted from different-sized convolution kernels are fused to obtain a new spectral feature map.

D. Global Local Convolution Transformer

Although SSFE captures the rich local information of HSI, MSSCFormer still lacks global information mining capability. To address this problem, we designed a GLCT module. We use the central pixel of $\mathbf{P}_i$ instead of the previous class token as the final classification vector. First, the long-range dependencies between the central pixel and the surrounding pixels are extracted by the SA mechanism. Then, we design an LCFB module to capture the multigranularity local feature information between pixels. This global local transformer structure is more conducive for the model to understand the spatial and spectral information of HSI.

In HSI classification tasks, we usually use the central pixel of the sample as the object to be classified. The spectral and spatial information of the central pixel and its surrounding pixels are analyzed comprehensively to determine the class of the central pixel. The overall structure of GLCT is similar to that of ViT [38], and the difference lies in the setting of the class token and the design of the MLP. In ViT, the class token is randomly initialized before model training, and then its relationship with the global image block is established, which finally serves as the final classified adjacency vector. GLCT uses the pixel at the center of the input data as a class token and establishes a connection with the global pixel through MHSA. After MHSA, the LCFB module is designed to model the local information of the image to make up for the shortcomings of ViT in image edges, texture, and other features. The overall structure of GLCT is shown in the right part of Fig. 1, which mainly consists of position encoding, MHSA, and LCFB. The calculation process can be expressed as follows:

$$\mathbf{T}_{\mathrm{in}} = [T_1, T_2, \ldots, T_{\mathrm{center}}, \ldots, T_N] + \mathrm{PE}_{\mathrm{pos}} \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

$$\mathbf{Z}_{\mathrm{GLCT}'} = \mathrm{MHSA}(\mathrm{LN}(\mathbf{T}_{\mathrm{in}})) + \mathbf{T}_{\mathrm{in}} \tag{12}$$

$$\mathbf{Z}_{\mathrm{GLCT}} = \mathrm{LCFB}(\mathrm{LN}(\mathbf{Z}_{\mathrm{GLCT}'})) + \mathbf{Z}_{\mathrm{GLCT}'} \tag{13}$$

where $\{T_{\mathrm{in}}\} \in \mathbb{R}^{N\times D}$, $N = H \times W$, $\mathrm{PE}_{\mathrm{pos}} \in \mathbb{R}^{\mathrm{N}\times\mathrm{D}}$, $\mathrm{LN}(\cdot)$ denotes the layer normalization function, and $\mathrm{LCFB}(\cdot)$ denotes the local context feature block.

The first part is the positional encoding. Inspired by SQSFormer [59], we use relative positional encoding instead of the commonly used absolute positional encoding, which can reduce the excessive noise caused by the absolute positional information and highlight the specificity of the central pixel position. Flatten the output of SSFE to get the representation of each token as follows:

$$\mathbf{T}_{\mathrm{in}} = [T_1, T_2, \ldots, T_{\mathrm{center}}, \ldots, T_N]. \tag{14}$$

The second part is the core component of Transformer, the multi-head SA (MHSA) mechanism, which is used to model the dependencies between different tokens. Compared with the single-head attention mechanism, MHSA has stronger feature modeling capabilities. This is because in MHSA, each attention head can independently learn different representations of the input data, so that it can capture diverse dependencies. For example, one attention head may focus on modeling grammatical features, while another may focus more on extracting semantic information. Specifically, the calculation process of MHSA is as follows:

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{15}$$

$$\mathbf{H}_i = \mathrm{Attention}\left(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V\right) \quad i = 1, 2, \ldots, h \tag{16}$$

$$\mathrm{MHSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Concat}(\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_h)\mathbf{W} \tag{17}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are three predefined learnable parameter matrices, $\mathrm{Softmax}(\cdot)$ is the activation function, $(d_k)^{1/2}$ is an important scaling factor used to scale the dot product results in the dot product attention mechanism to avoid the problem of gradient vanishing in the softmax function, h is the number of heads, $\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V$ are the parameter matrices corresponding to the header, where $W \in \mathbb{R}^{d_{\mathrm{models}}\times hd_v}$, d_{models} is the dimensionality of the input embedding, and the values of d_k and d_v are equal to (d_{models}/h). We regard this part as a modeling method of global spectral information from a global perspective.

The third part is the LCFB module. In the traditional Transformer, the MHSA mechanism performs well in capturing global image information. However, the Transformer has certain limitations in capturing local information. In order to make up for the shortcomings of the Transformer in capturing local information, as shown in the middle part of Fig. 1, we design an LCFB module. Specifically, LCFB consists of three convolution blocks. Conv1 and Conv3 consist of BN layers, GELU activation functions, and 2-D convolutions with a kernel size of 1. BN and GELU activation functions enable convolution operations to be performed under stable input conditions. 2-D convolutions are used to dynamically adjust the number of channels of the feature map, which is convenient for enhancing feature expressiveness while keeping the image space size unchanged. The Conv2 convolution part consists of a BN layer, a GELU activation function, and two 2-D deep convolutions with kernel sizes of 3 and 7, respectively, which are used to extract multigranularity local features of the image. This multibranch design enables the model to perceive both fine and coarse textures within local regions, which is crucial for HSI classification where subtle spatial variations carry significant semantic information. Compared with existing ViT variants that may adopt simple local convolutions for auxiliary enhancement, the proposed LCFB integrates multiscale convolutional paths in a residual manner, forming a hierarchical local representation enhancement module. This structure allows not only for multigranularity spatial context aggregation but also strengthens spatial continuity modeling that MHSA alone lacks. Finally, the residual connection is used to add the convolution result to the input, which can not only maintain the original spectral feature information but also fuse the local feature information after convolution. Experiments show that the multigranularity feature extraction effect of 2-D deep convolution with kernel sizes of 3 and 7 is the best. Finally, the output vector of the transformer is further mapped to a specific class space through a linear classifier. Obviously, LCFB is a method for mining local details of HSI from a local perspective.

The "global guides local, local corrects global" iterative mechanism of GLCT goes beyond the unidirectional global modeling of ViT and is more adapted to the complex spatial–spectral coupling characteristics of HSI.

E. Classification MLP

To achieve accurate and efficient classification, we employ a multilayer perceptron (MLP), which consists of two fully connected layers, a BN layer, a Dropout layer, and a ReLU activation function. In this article, the output classification token of GLCT is used as the input of the classification MLP. The output dimension of the last fully connected layer is the number of land classes to be predicted in the image, and the highest value among all the output classes corresponds to the final predicted label. The entire execution of the proposed method is outlined in Algorithm 1.

IV. Experiments and Analysis

A. Dataset Description

1) Houston2013: The Houston2013 (HS) is acquired over the University of Houston campus and the neighboring urban area. This scene consists of 349 × 1905 pixels and has 144 spectral bands in the wavelength range of 380–1050 nm. The spatial resolution is 2.5 m. For classification purposes, 15 land-cover classes are defined for this dataset. In the experiment, we randomly take ten samples for each class as training samples and the rest as test samples. Fig. 2 demonstrates the false-color map, the ground-truth map, and the 15 different land-cover color class labels in the HS dataset. The names of the land-cover classes and the sample details are shown in Table I.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Procedure of MSSCFormer

Input: HSI data $\mathbf{I} \in \mathbb{R}^{H\times W\times B}$ and label $\mathbf{Y} \in \mathbb{R}^{H\times W}$; PCA bands number b; patch size p.

Output: Predicted labels of the test set $\mathbf{Y}$.

Obtain the I_{pca} after PCA operation.
Obtain HSI patches from I_{pca} and split them into trainset and testset.
Set batch size = 64; pca = 10; learning rate = 0.001; epochs = 50.
for $i = 1$ to epochs **do**
  Perform ICSA module on patches to reassign spectral weights for each sample P_i'.
  P_i' is used as the input of SSFE to extract shallow spatial features and multi-granularity spectral features F_{ssfe}.
  Reshape F_{ssfe} into a 2-D feature map, and then flatten the 2-D data into a 1-D vector F.
  Obtain T_{in} by adding positional information. In this article, we use the center token of the T_{in} as the classification token T_c.
  Feed T_{in} into the GLCT for feature learning.
  for $j = 1$ to L **do**
    Perform MHSA operation by Eq. (12).
    Perform LCFB operation by Eq. (13).
  end for
  Input T_c to the classification mlp to obtain the class label.
end for
Predict the corresponding labels of testset.

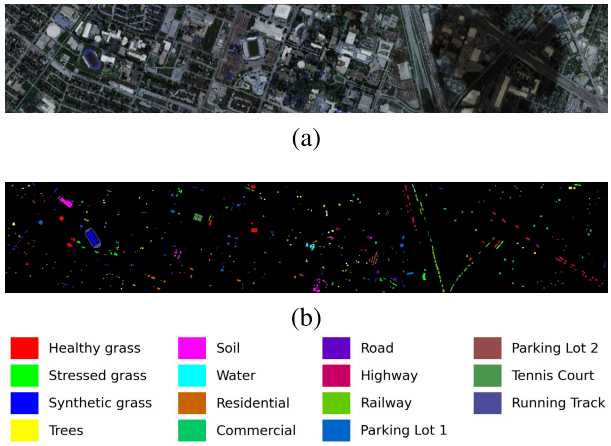


Fig. 2. Houston2013 dataset. (a) False-color map. (b) Ground-truth map.

2) Dioni Dataset: The Dioni dataset (Dioni) is from the HyRANK benchmark, which is acquired by the Hyperion sensor on the National Aeronautics and Space Administration (NASA) Earth Observation 1 (EO-1) satellite. This dataset has 176 spectral bands and a spatial size of 250 × 1376. There are 12 land-cover classes in the Dioni dataset. Fig. 3 demonstrates the false-color map, the ground-truth map, and the 12 different land-cover color class labels in the Dioni dataset. The names of the land-cover classes and the sample details are shown in Table II.

TABLE I

DETAILS OF CLASSES IN HOUSTON2013 DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train	Test
C01	Healthy grass	10	1241
C02	Stressed grass	10	1244
C03	Synthetic grass	10	687
C04	Trees	10	1234
C05	Soil	10	1232
C06	Water	10	315
C07	Residential	10	1258
C08	Commercial	10	1234
C09	Road	10	1242
C10	Highway	10	1217
C11	Railway	10	1225
C12	Parking Lot 1	10	1223
C13	Parking Lot 2	10	459
C14	Tennis Court	10	418
C15	Running Track	10	650
	TOTAL	150	14879

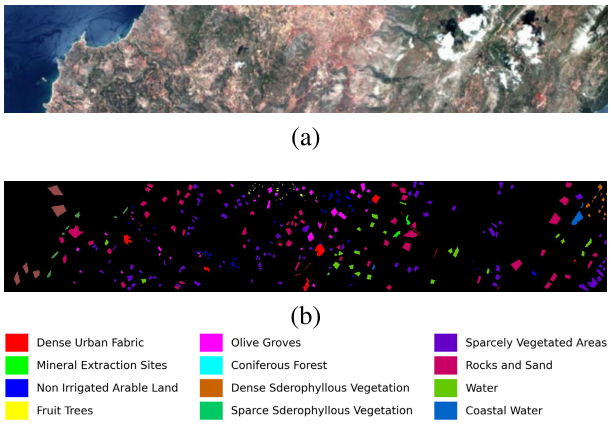


Fig. 3. Dioni dataset. (a) False-color map. (b) Ground-truth map.

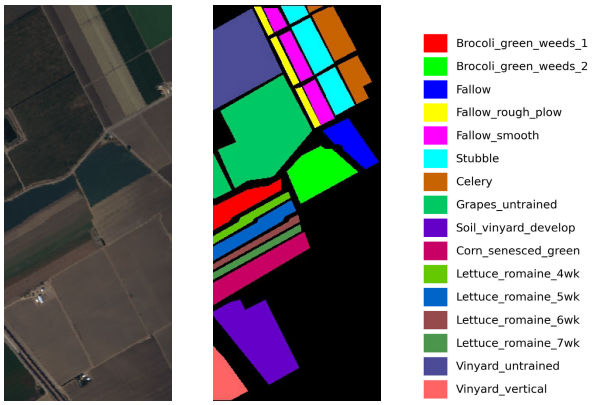
TABLE II

DETAILS OF CLASSES IN DIONI DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train	Test
C01	Dense Urban Fabric	20	1242
C02	Mineral Extraction Sites	20	184
C03	Non Irrigated Arable Land	20	594
C04	Fruit Trees	20	130
C05	Olive Groves	20	1748
C06	Coniferous Forest	20	341
C07	Dense Sderophyllous Vegetation	20	5015
C08	Sparce Sderophyllous Vegetation	20	6354
C09	Sparcely Vegetated Areas	20	1734
C10	Rocks and Sand	20	472
C11	Water	20	1592
C12	Coastal Water	20	378
	TOTAL	240	19784

3) Salinas Dataset: The Salinas dataset (SA) was collected over the Salinas Valley, CA, USA. It features a spatial resolution of 3.7 m and has dimensions of 512 × 217 × 204, where 512 and 217 represent the pixel count in the height and width dimensions, and 204 denotes the number of spectral bands. Fig. 4 demonstrates the false-color map, the ground-truth map, and the 16 different land-cover color class labels in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. Salinas dataset. (a) False-color map. (b) Ground-truth map.

TABLE III

DETAILS OF CLASSES IN SALINAS DATASET AND THE NUMBER OF SAMPLES USED FOR TRAINING AND TESTING, RESPECTIVELY

No.	Class	Train	Test
C01	Brocoli green weeds1	20	1989
C02	Brocoli green weeds2	20	3706
C03	Fallow	20	1956
C04	Fallow rough plow	20	1374
C05	Fallow smooth	20	2658
C06	Stubble	20	3939
C07	Celery	20	3559
C08	Grapes untrained	20	11251
C09	Soil vinyard develop	20	6183
C10	Corn senesced green	20	3258
C11	Lettuce romaine 4wk	20	1048
C12	Lettuce romaine 5wk	20	1907
C13	Lettuce romaine 6wk	20	896
C14	Lettuce romaine 7wk	20	1050
C15	Vinyard untrained	20	7248
C16	Vinyard vertical trellis	20	1787
	TOTAL	320	53809

the Salinas dataset. The names of the land-cover classes and the sample details are shown in Table III.

B. Experimental Setup

1) Evaluation Metrics: In order to measure the performance of the proposed method and other comparative methods, we use regular classification evaluation metrics for quantitative analysis, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient (k). Larger values of each metric indicate better classification performance.

2) Comparison Methods: In our performance comparison experiments, we extensively compare the proposed method with the current state-of-the-art and classical methods. These methods can be classified into three classes. CNN-based methods including FDSSC [60], HybridSN [61], and DBDA [43]. Transformer-based methods including ViT [38], SpectralFormer [48], and MFT [62]. Methods based on the mixture of CNN and Transformer, including HybridFormer [57], MASSFormer [58], and DCTN [41].

1) FDSSC [60] first uses three 3-D convolutions to simultaneously extract information in the spatial and spectral dimensions. Then, it uses a 2-D convolution layer to further process the spatial information and uses dense connections to fuse the spectral and spatial features.
2) HybridSN [61] combines 2-D convolution and 3-D convolution, fully demonstrating the advantages of CNNs in HSI classification tasks.
3) DBDA [43] uses a dual-branch dual attention structure to extract the spatial and spectral features of the image, respectively. Finally, the features output by the two branches are fused to obtain the final feature map.
4) ViT [38] divides the input image into multiple patches of equal size, applies positional encoding, and feeds them into a sequence of Transformer encoders to capture the global spectral information of the image.
5) SpectralFormer [48] uses groupwise spectral embedding to extract spectral information of adjacent bands and uses cross-layer adaptive fusion to combine shallow and deep spectral features.
6) MFT [62] combines data from different modes and utilizes specialized modal feature extraction, a Transformer encoder, and a multimodal attention mechanism for modal feature extraction, fusion, and global modeling.
7) HybridFormer [57] uses a multigranularity feature learning method to capture important spatial and spectral information from different scales. In addition, the spatial–spectral attention mechanism is used to enhance the focus on important regions and bands in the image.
8) MASSFormer [58] incorporates a memory tokenizer and a memory-augmented transformer encoder, which enhances information fusion with prior knowledge while mitigating the deep decay problem of models.
9) DCTN [41] adopts the spatial–spectral fusion projection module to effectively capture spatial–spectral features in parallel. Furthermore, it uses a Transformer with efficient interactive SA to fuse features from three different dimensions.

3) Implementation Details: For our MSSCFormer, we use Adam optimizer with a learning rate of 0.001 to guide training. The batch size is set to 64. The training epoch is set to 50, and validation is performed every ten rounds. The PCA size is uniformly set to 10. In the Transformer architecture, the embedding dimension is set to 64 to effectively represent the input data. In addition, we conduct a detailed analysis of some important parameters to meet the characteristics of the dataset, including the size of the input data patch, the number of Transformer blocks, and the number of attention heads. For the patch size, the best parameters for HS, Dioni, and SA are 13, 17, and 17, respectively. For Transformer, the number of Transformer encoders and the number of attention heads used in them have a great impact on the performance of the model. Experiments show that the best parameters for HS, Dioni, and SA in terms of the number of Transformer encoders are 2,3, and 3, respectively. For the number of attention heads, the best parameters for HS, Dioni, and SA are 8, 20, and 8, respectively.

All experiments were performed 20 times, and the average of the 20 times was taken as the final result. We use the PyTorch training framework for model implementation and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

training. The devices used are an Intel Core i5-13400F CPU and an NVIDIA RTX 3090 24 G GPU.

C. Parameter Analysis

In order to conduct a comprehensive and in-depth study of the proposed MSSCFormer model, we conducted a detailed analysis of some important hyperparameters in the experiment. These hyperparameters include the size of the input image patch, the number of Transformer encoders, and the number of attention heads. The size of these hyperparameters directly affects the complexity of the model and has an important impact on the classification performance of the model. When analyzing each hyperparameter, the other experimental parameters are set according to Section IV-B.

1) Impact of the Number of Transformer Encoders: The number of Transformer encoders has a significant impact on its global modeling ability of the data. In our experiments, blocks of size 1, 2, 3, 4, and 5 were considered to test its impact on the classification accuracy of the model. Fig. 5(a) depicts the changes in the evaluation index OA of the three datasets corresponding to different sizes of Transformer encoders. The HS dataset achieved the best classification effect when the number of Transformer encoders was 2, while the Dioni and SA datasets achieved the best classification effect when the number of Transformer encoders was 3. Therefore, we set the Transformer encoders of the HS dataset to 2 and the Transformer encoders of the Dioni and SA datasets to 3.

2) Impact of the Number of Attention Heads in MHSA: The number of attention heads in the Transformer's MHSA affects the expressiveness and complexity of the model, so it is necessary to choose the right number of attention heads. In order to find the best number of attention heads, we selected a set of sizes of 4, 6, 8, 10, 15, 20, 25, and 30. Fig. 5(b) depicts the changes in the evaluation index OA of the three datasets corresponding to different sizes of attention heads. The HS and SA datasets achieved the best classification performance when the number of attention heads was 8, while the Dioni dataset achieved the best classification performance when the number of attention heads was 20. Therefore, we set the attention heads corresponding to the HS and SA datasets to 8, and the attention heads of the Dioni dataset to 20.

3) Impact of Different Patch Size: The size of the patch directly affects the processing of model input data. In order to analyze the impact of the patch on model classification performance, we consider five values from small to large, 5, 9, 13, 17, and 21, while keeping other hyperparameters unchanged. The five values in this range can well reflect the influence of different patch sizes on the model. Fig. 5(c) depicts the changes of evaluation index OA corresponding to patches of different sizes in the three datasets. As can be seen from the figure, the HS dataset can achieve the best classification effect when the patch is 13, while Dioni and SA achieve the best classification effect when the patch is 17. Therefore, we set the patch of HS to 13, and the patch of Dioni and SA to 17.

TABLE IV
ABLATION STUDY RESULTS OF THE PROPOSED COMPONENTS ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Cases	Component			Dataset		
	ICSA	SSFE	LCFB	HS	Dioni	SA
1	✗	✗	✗	84.06	84.73	91.49
2	✓	✗	✗	84.43	84.93	92.41
3	✗	✓	✗	86.16	86.51	93.34
4	✗	✗	✓	84.53	84.82	93.22
5	✓	✓	✗	86.68	86.82	93.56
6	✗	✓	✓	86.51	86.84	94.26
7	✓	✓	✓	**87.26**	**87.40**	**94.35**

TABLE V
ABLATION STUDY RESULTS OF ICSA ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

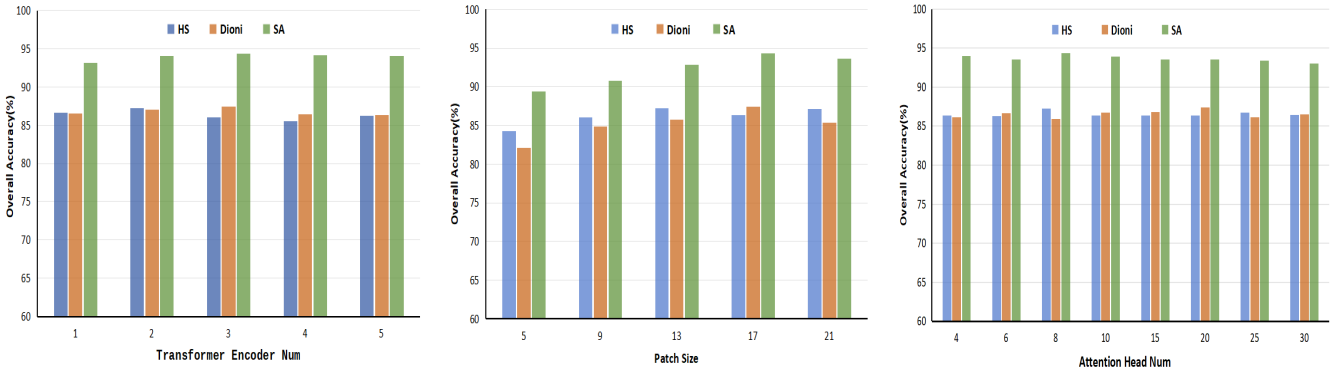
Cases	Component				Dataset		
	SE	ECA	SA	ICSA	HS	Dioni	SA
1	✓	✗	✗	✗	84.67	86.20	93.47
2	✗	✓	✗	✗	85.61	86.24	93.74
3	✗	✗	✓	✗	86.07	86.05	93.68
4	✗	✗	✗	✓	**87.26**	**87.40**	**94.35**

D. Ablation Studies

1) Ablation Experiments on the Main Modules: To analyze the contribution of different modules in the MSSCFormer to the model performance, we designed comprehensive ablation experiments on three datasets. MSSCFormer consists of three main modules: ICSA, SSFE, and LCFB. In addition, we tested the impact of different combinations of convolutions in LCFB on the model performance.

We conducted a comprehensive analysis of the ablation experiments, using seven combinations of ablation in three modules. Table IV records the impact of the seven ablation combinations on the model classification performance from the evaluation indicator OA. Specifically, compared with Case 1, in Cases 2, 3, and 4, the OA improves when each module is used separately, suggesting that each module plays a positive role. In addition, we found excellent compatibility between the modules; two-by-two combination of modules worked better than using a single module alone. In Case 5, ICSA narrows the intra-class spectral differences and filters out redundant spectral information, alleviating SSFE from extracting redundant feature information and allowing SSFE to extract more discriminative feature information, so Case 5 is more effective than Case 2 and Case 3 when used separately. Furthermore, in Case 6, SSFE extracts multigranularity features while maintaining a low computational complexity, and LCFB compensates for Transformer's lack of local information extraction ability, so it is more advantageous than Case 3, which only uses the Transformer to capture global spectral information. In Case 7, the best results are achieved when the three modules are used together, with improved accuracies of 3.2%, 2.67%, and 2.77% on the three datasets, respectively. Overall, each module served a positive role when used separately and demonstrated excellent complementarity when used together.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 5. Impact of select hyperparameters on OA of three datasets (HS, Dioni, SA). (a) Impact of the number of transformer encoders. (b) Impact of the number of attention heads in MHSA. (c) Impact of different patch sizes.

2) Ablation Study on the ICSA Module: To better analyze why ICSA is superior to existing attention-based spectral normalization, we conducted several spectral attention experiments and added theoretical analysis to validate the advantages of ICSA. Table V shows the result of different spectral attention mechanisms. ICSA achieved the best results on all three datasets. ICSA outperforms the second precision method by 1.19%, 1.16%, and 0.64% on the HS, Dioni, and SA datasets, respectively. The main reasons for low results when using SE are that our experiments employ a small number of training samples, the statistics computed by global pooling are susceptible to single-sample outliers or noise, which leads to large fluctuations in the learned channel weights under different training randomness. The reason ICSA achieves superior performance compared to traditional attention mechanisms such as SE, ECA, and SA is that these methods focus primarily on enhancing discriminative features by learning global or local channel-wise importance but ignore the intrinsic spectral variability among samples of the same class in HSI. In contrast, ICSA is specifically designed to address this issue by incorporating a learnable class-center constraint, which effectively reduces intra-class spectral differences through an adaptive learning strategy.

3) Ablation Study on the MGSFF Module: To verify the effectiveness of four-way grouped convolution in MGSFF, we designed ablation experiments with different convolutional grouped methods, including single-branch, two-branch, three-branch, four-branch, five-branch, and six-branch. We did several experiments on the single-branch method to verify the necessity of grouping. Table VI records the results of different grouped convolution methods. It can be clearly seen that the four-branch grouped convolution method works best while keeping the dimensionality constant. One possible reason is that single-branch or few-group convolution is prone to bias toward a specific scale, resulting in poor adaptability to objects with different spectral patterns and lacking the ability to capture both local and global spectral context. Another possible reason is that excessive group convolution makes it difficult for the model to extract rich patterns from limited dimensions, especially in complex feature extraction.

4) Ablation Study on the LCFB Module: To further validate the impact of LCFB on the model performance, we compared

TABLE VI

ABLATION STUDY RESULTS OF MGSFF ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Cases	Component						Dataset		
	Conv3 (kernel=3)	Conv9 (kernel=9)	Conv15 (kernel=15)	Conv21 (kernel=21)	Conv25 (kernel=25)	Conv31 (kernel=31)	HS	Dioni	SA
1	✓	✗	✗	✗	✗	✗	86.49	86.56	93.46
2	✗	✓	✗	✗	✗	✗	85.79	86.73	93.42
3	✗	✗	✓	✗	✗	✗	86.34	86.29	93.61
4	✗	✗	✗	✓	✗	✗	86.33	86.85	93.55
5	✓	✓	✗	✗	✗	✗	86.59	86.47	93.60
6	✓	✓	✓	✗	✗	✗	86.25	86.51	93.34
7	✓	✓	✓	✓	✗	✗	**87.26**	**87.40**	**94.35**
8	✓	✓	✓	✓	✓	✗	86.32	86.85	93.45
9	✓	✓	✓	✓	✓	✓	86.07	86.31	93.61

TABLE VII

ABLATION STUDY RESULTS OF LCFB ON DIFFERENT DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Cases	Component				Dataset		
	Conv3 (kernel=3)	Conv5 (kernel=5)	Conv7 (kernel=7)	Conv9 (kernel=9)	HS	Dioni	SA
1	✗	✗	✗	✗	86.68	86.82	93.36
2	✓	✗	✗	✗	87.19	87.13	94.03
3	✗	✓	✗	✗	87.11	86.88	94.21
4	✗	✗	✓	✗	87.01	87.05	94.19
5	✗	✗	✗	✓	86.98	86.84	94.08
6	✓	✓	✗	✗	87.22	86.96	94.05
7	✓	✗	✓	✗	**87.26**	**87.40**	**94.35**
8	✓	✗	✗	✓	87.04	86.92	93.92
9	✗	✓	✓	✗	86.87	86.99	94.08
10	✗	✓	✗	✓	87.01	87.17	93.53

different combinations of convolutions on the ability of LCFB to extract local information in the Transformer. Only the combinations of different convolutions in the LCFB were modified while maintaining the rest of the network structure. Table VII records the effect of nine different convolution combinations on the classification accuracy of the model. It can be seen that for all three datasets, features are extracted best when using convolutions with kernel sizes 3 and 7 as a two-branch convolutional structure. The accuracy of the model on the HS, Dioni, and SA datasets increases by 0.58%, 0.58%, and 0.79%, respectively, when MSSCFormer uses them compared to when they are missing.

E. Classification Results and Analysis

To validate the superiority of the proposed MSSCFormer model, we implemented nine representative algorithms for comparison. We divide them into three groups: CNN-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET (THE BEST RESULT ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	FDSSC	HybridSN	DBDA	ViT	SpectralFormer	MFT	HybridFormer	MASSFormer	DCTN	MSSCFormer (Ours)
Healthy grass	89.06 ± 2.63	**92.45 ± 4.17**	89.85 ± 2.94	89.80 ± 1.47	89.92 ± 4.19	89.63 ± 2.08	91.24 ± 3.12	92.42 ± 1.69	90.37 ± 2.70	90.96 ± 2.40
Stressed grass	89.94 ± 1.91	91.63 ± 2.99	90.09 ± 1.48	91.31 ± 0.70	90.18 ± 2.32	88.90 ± 2.24	92.08 ± 2.55	**93.64 ± 1.98**	90.70 ± 1.39	91.76 ± 2.56
Synthetic grass	99.54 ± 0.60	99.35 ± 0.63	99.51 ± 0.47	99.87 ± 0.12	99.94 ± 0.11	99.76 ± 0.27	**99.97 ± 0.05**	99.96 ± 0.08	99.95 ± 0.06	99.69 ± 1.33
Trees	76.97 ± 6.74	80.73 ± 9.09	79.89 ± 3.19	83.65 ± 2.70	86.64 ± 3.75	78.78 ± 3.05	85.06 ± 4.04	85.61 ± 3.18	80.25 ± 3.09	**88.32 ± 3.36**
Soil	99.46 ± 0.82	98.59 ± 9.21	99.31 ± 0.83	98.68 ± 1.36	95.78 ± 7.39	88.45 ± 8.66	96.53 ± 9.71	98.54 ± 3.69	98.55 ± 3.15	**99.61 ± 1.06**
Water	88.17 ± 0.59	95.09 ± 4.82	88.66 ± 0.43	88.69 ± 0.33	**97.26 ± 2.42**	90.80 ± 3.97	88.46 ± 1.01	96.57 ± 3.36	91.93 ± 3.03	93.31 ± 3.68
Residential	75.97 ± 11.37	61.96 ± 11.17	75.51 ± 5.61	62.54 ± 3.60	67.30 ± 6.75	61.21 ± 7.09	73.96 ± 4.32	82.53 ± 5.50	67.87 ± 5.97	**90.93 ± 3.51**
Commercial	56.01 ± 7.23	63.67 ± 7.76	56.64 ± 4.06	68.24 ± 4.46	58.23 ± 4.77	52.32 ± 8.72	64.60 ± 4.94	67.20 ± 4.55	60.48 ± 6.60	**70.14 ± 6.84**
Road	54.61 ± 8.77	65.13 ± 6.67	65.12 ± 5.54	46.56 ± 4.00	66.14 ± 7.39	57.13 ± 8.59	58.27 ± 11.19	71.85 ± 4.24	52.59 ± 8.54	**76.03 ± 7.52**
Highway	86.10 ± 6.43	79.56 ± 11.63	84.56 ± 2.87	80.04 ± 3.11	81.93 ± 7.56	81.35 ± 8.45	82.74 ± 5.83	83.71 ± 3.25	88.68 ± 5.05	**90.00 ± 5.17**
Railway	**77.87 ± 7.32**	68.37 ± 1.73	68.47 ± 2.72	69.11 ± 2.00	67.98 ± 3.71	71.15 ± 6.06	74.92 ± 5.22	69.23 ± 1.87	70.90 ± 6.68	67.87 ± 1.35
Parking Lot 1	64.39 ± 11.89	74.10 ± 14.18	70.18 ± 4.09	70.49 ± 2.51	65.83 ± 7.58	68.36 ± 8.00	66.54 ± 6.81	78.32 ± 4.15	67.14 ± 5.74	**85.08 ± 3.49**
Parking Lot 2	90.90 ± 4.86	88.85 ± 9.97	92.22 ± 2.19	91.35 ± 2.34	88.12 ± 4.74	90.42 ± 3.37	91.63 ± 2.32	**93.78 ± 1.24**	88.32 ± 3.39	93.25 ± 2.07
Tennis Court	99.97 ± 0.11	99.18 ± 2.01	100 ± 0.00	100.00 ± 0.00	99.17 ± 0.88	100.00 ± 0.00	99.89 ± 0.42	100.00 ± 0.00	100.00 ± 0.00	**100.00 ± 0.00**
Running Track	98.90 ± 1.72	98.98 ± 2.20	99.43 ± 1.30	99.04 ± 0.79	99.83 ± 0.19	97.30 ± 3.53	**99.99 ± 0.03**	97.42 ± 1.24	99.60 ± 0.55	99.96 ± 0.13
OA	80.32 ± 1.93	80.87 ± 3.16	81.18 ± 0.47	79.54 ± 0.46	80.44 ± 2.24	77.56 ± 2.55	81.47 ± 3.06	84.95 ± 0.86	80.14 ± 5.33	**87.26 ± 0.87**
AA	80.20 ± 2.17	81.13 ± 3.04	81.04 ± 0.44	79.36 ± 0.74	80.25 ± 1.00	77.07 ± 2.05	81.50 ± 0.80	84.77 ± 0.75	79.99 ± 4.37	**87.55 ± 0.27**
$k \times 100$	80.45 ± 1.66	80.6 ± 3.14	81.33 ± 0.46	79.72 ± 0.11	80.64 ± 3.41	78.04 ± 2.57	81.45 ± 5.31	85.12 ± 0.91	80.28 ± 6.25	**86.96 ± 1.31**

TABLE IX

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DIONI DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	FDSSC	HybridSN	DBDA	ViT	SpectralFormer	MFT	HybridFormer	MASSFormer	DCTN	MSSCFormer (Ours)
Dense Urban Fabric	66.78 ± 1.58	77.96 ± 4.27	68.09 ± 1.03	72.37 ± 3.05	74.38 ± 4.26	82.52 ± 6.26	75.31 ± 3.96	83.34 ± 3.48	73.97 ± 1.87	**84.16 ± 1.58**
Mineral Extraction Sites	99.42 ± 2.55	99.75 ± 0.54	100.00 ± 0.00	99.72 ± 0.69	99.75 ± 0.59	100.00 ± 0.00	99.24 ± 0.24	99.64 ± 0.98	100.00 ± 0.00	**100.00 ± 0.00**
Non Irrigated Arable Land	85.89 ± 1.22	85.05 ± 4.44	86.29 ± 1.04	85.21 ± 1.20	85.36 ± 4.09	86.16 ± 1.40	87.41 ± 1.54	**88.35 ± 3.98**	85.94 ± 0.64	87.77 ± 3.13
Fruit Trees	94.11 ± 1.03	93.50 ± 1.04	94.00 ± 0.68	94.42 ± 0.78	94.42 ± 1.26	**95.73 ± 1.28**	94.88 ± 1.43	93.84 ± 1.34	94.30 ± 1.12	93.65 ± 0.55
Olive Groves	74.05 ± 4.53	73.92 ± 5.33	74.10 ± 2.24	73.43 ± 2.68	77.46 ± 4.14	75.14 ± 3.11	77.26 ± 1.46	76.65 ± 2.93	73.93 ± 2.52	**79.78 ± 1.93**
Coniferous Forest	100.00 ± 0.00	99.97 ± 0.09	100.00 ± 0.00	99.75 ± 0.72	99.88 ± 0.40	99.82 ± 0.53	99.88 ± 0.29	99.92 ± 0.23	**100.00 ± 0.00**	99.97 ± 0.09
Dense Sderophyllous Vegetation	75.12 ± 8.30	85.98 ± 2.56	77.93 ± 2.28	81.14 ± 1.67	78.35 ± 3.40	74.14 ± 5.33	80.07 ± 2.96	85.72 ± 2.79	76.08 ± 3.21	**86.93 ± 1.41**
Sparce Sderophyllous Vegetation	74.05 ± 15.49	67.55 ± 5.89	77.97 ± 2.38	71.94 ± 2.94	70.41 ± 5.78	77.77 ± 6.27	73.79 ± 3.44	72.09 ± 5.90	72.45 ± 4.41	**83.31 ± 3.29**
Sparcely Vegetated Areas	91.74 ± 3.27	89.72 ± 3.52	**92.94 ± 1.36**	89.28 ± 2.36	92.12 ± 2.27	91.76 ± 2.88	91.03 ± 3.24	91.49 ± 2.29	92.42 ± 2.38	92.29 ± 1.67
Rocks and Sand	98.83 ± 1.23	97.09 ± 3.48	99.15 ± 0.00	98.24 ± 0.71	98.29 ± 1.29	**99.15 ± 0.00**	99.08 ± 0.15	98.80 ± 1.22	99.14 ± 0.04	99.02 ± 0.31
Water	99.99 ± 0.01	99.38 ± 1.32	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	**100.00 ± 0.00**
Coastal Water	99.96 ± 0.17	93.84 ± 10.73	**99.98 ± 0.05**	99.96 ± 0.12	99.52 ± 0.98	99.28 ± 1.81	99.21 ± 2.42	98.62 ± 1.41	99.96 ± 0.12	98.08 ± 1.74
OA	79.76 ± 11.09	80.70 ± 3.25	81.95 ± 0.79	80.65 ± 1.15	80.19 ± 4.73	81.81 ± 4.25	81.73 ± 1.33	82.96 ± 2.75	80.01 ± 2.37	**87.40 ± 1.17**
AA	79.10 ± 11.98	81.52 ± 0.42	81.78 ± 0.53	80.83 ± 1.54	79.68 ± 6.02	81.75 ± 7.01	82.07 ± 1.06	83.00 ± 2.92	80.67 ± 1.76	**87.51 ± 0.83**
$k \times 100$	80.42 ± 9.33	79.88 ± 4.73	82.12 ± 1.00	80.48 ± 0.69	80.70 ± 2.91	81.87 ± 1.48	81.40 ± 1.37	82.92 ± 2.57	79.33 ± 2.08	**87.29 ± 1.47**

TABLE X

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SALINAS DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Class	CNN-based Methods			Transformer-based Methods			CNN-Transformer-based Methods			
	FDSSC	HybridSN	DBDA	ViT	SpectralFormer	MFT	HybridFormer	MASSFormer	DCTN	MSSCFormer (Ours)
Brocoli green weeds1	99.99 ± 0.02	99.84 ± 0.44	100.00 ± 0.00	99.83 ± 0.60	99.87 ± 0.16	99.82 ± 0.19	99.73 ± 0.37	99.97 ± 0.04	99.83 ± 0.38	**100.00 ± 0.00**
Brocoli green weeds2	100.00 ± 0.00	99.91 ± 0.33	100.00 ± 0.00	99.98 ± 0.07	99.97 ± 0.04	99.82 ± 0.38	99.79 ± 0.33	**100.00 ± 0.00**	98.97 ± 3.04	99.96 ± 0.16
Fallow	99.59 ± 1.80	99.61 ± 1.13	99.96 ± 0.07	99.68 ± 0.80	99.87 ± 0.20	99.46 ± 0.78	99.53 ± 0.56	99.94 ± 0.09	99.78 ± 0.51	**99.97 ± 0.05**
Fallow rough plow	**99.69 ± 0.35**	97.62 ± 1.55	98.91 ± 0.55	97.30 ± 0.91	97.96 ± 2.00	94.30 ± 3.99	98.02 ± 1.83	98.24 ± 0.52	97.93 ± 2.91	98.82 ± 1.10
Fallow smooth	**99.01 ± 0.53**	96.40 ± 2.88	96.34 ± 1.03	97.50 ± 0.79	98.25 ± 1.58	89.63 ± 3.67	97.45 ± 2.06	98.05 ± 0.68	95.38 ± 2.24	98.78 ± 0.55
Stubble	**99.99 ± 0.01**	99.23 ± 1.34	99.99 ± 0.01	99.72 ± 0.25	99.92 ± 0.11	99.77 ± 0.28	99.68 ± 0.37	99.86 ± 0.31	99.61 ± 1.24	99.70 ± 0.24
Celery	99.99 ± 0.01	99.65 ± 0.44	99.85 ± 0.10	99.80 ± 0.14	99.80 ± 0.13	99.23 ± 0.66	99.41 ± 0.96	99.98 ± 0.04	99.81 ± 0.23	**99.99 ± 0.01**
Grapes untrained	81.11 ± 12.50	81.13 ± 11.06	74.42 ± 12.58	67.45 ± 9.87	62.37 ± 8.14	78.91 ± 12.80	73.36 ± 9.79	79.59 ± 2.80	82.24 ± 7.97	**84.11 ± 7.59**
Soil vinyard develop	99.78 ± 0.36	99.73 ± 0.68	**99.92 ± 0.17**	99.49 ± 1.10	99.70 ± 0.63	99.64 ± 0.67	97.37 ± 4.64	99.78 ± 0.50	97.22 ± 4.75	99.80 ± 0.45
Corn senesced green	97.21 ± 0.74	97.06 ± 1.34	97.92 ± 0.39	96.61 ± 1.13	96.64 ± 0.66	97.41 ± 1.05	96.49 ± 1.16	98.09 ± 0.47	96.97 ± 1.02	**98.23 ± 0.63**
Lettuce romaine 4wk	**99.93 ± 0.29**	99.56 ± 0.70	99.87 ± 0.24	99.41 ± 1.62	99.44 ± 0.60	97.10 ± 2.09	98.77 ± 1.23	99.73 ± 0.44	98.22 ± 1.18	99.16 ± 0.98
Lettuce romaine 5wk	99.92 ± 0.10	97.81 ± 2.49	99.67 ± 0.29	98.59 ± 1.35	**99.88 ± 0.21**	97.88 ± 2.25	97.93 ± 2.40	99.19 ± 0.66	98.50 ± 2.65	99.38 ± 0.65
Lettuce romaine 6wk	99.32 ± 0.75	97.63 ± 5.07	99.38 ± 0.58	99.15 ± 1.34	99.07 ± 0.89	98.81 ± 1.19	97.79 ± 2.18	**99.87 ± 0.49**	98.99 ± 1.01	98.95 ± 1.82
Lettuce romaine 7wk	100.00 ± 0.00	99.56 ± 0.54	**100.00 ± 0.00**	99.93 ± 0.09	99.94 ± 0.06	99.19 ± 1.13	99.11 ± 1.38	99.99 ± 0.04	99.26 ± 1.77	99.51 ± 0.36
Vinyard untrained	80.62 ± 11.93	86.44 ± 13.53	82.67 ± 10.65	80.64 ± 7.43	83.73 ± 7.45	80.00 ± 17.40	83.42 ± 13.25	**89.04 ± 3.11**	86.47 ± 6.91	85.12 ± 10.58
Vinyard vertical trellis	99.39 ± 0.15	97.99 ± 5.37	99.18 ± 0.18	98.74 ± 0.90	98.75 ± 1.13	99.14 ± 0.92	98.84 ± 1.27	99.35 ± 0.35	99.61 ± 0.14	**99.67 ± 0.26**
OA	93.14 ± 2.70	93.47 ± 2.86	91.91 ± 2.61	89.96 ± 1.49	89.46 ± 0.82	91.74 ± 3.54	91.21 ± 3.76	93.90 ± 0.48	93.43 ± 1.49	**94.35 ± 1.30**
AA	92.58 ± 4.25	93.13 ± 3.59	92.16 ± 0.80	89.49 ± 1.54	89.77 ± 0.91	92.33 ± 3.36	90.55 ± 4.00	94.01 ± 0.36	93.63 ± 1.16	**94.31 ± 1.57**
$k \times 100$	93.69 ± 0.52	93.82 ± 1.88	91.66 ± 4.29	90.43 ± 1.00	89.15 ± 0.54	91.14 ± 3.02	91.88 ± 2.63	93.79 ± 0.58	93.22 ± 1.73	**94.39 ± 1.03**

TABLE XI

FLOPS AND NUMBER OF PARAMETERS OF DIFFERENT METHODS ON THREE DATASETS

Methods		FLOPs (M)			Parameters (M)		
		HS	Dioni	SA	HS	Dioni	SA
CNN-based methods	FDSSC	259.44	545.48	634.57	0.85	1.04	1.21
	HybridSN	940.23	1964.05	2275.75	5.48	8.03	8.55
	DBDA	161.14	338.36	393.22	0.28	0.34	0.39
Transformer-based methods	ViT	108.56	460.46	281.51	0.64	1.54	0.96
	SpectralFormer	42.09	185.89	95.12	0.29	1.09	0.52
	MFT	161.14	72.99	84.97	0.31	0.34	0.36
CNN-Transformer-based methods	HybridFormer	120.31	333.22	399.60	1.92	4.07	3.20
	MASSF	28.18	204.31	86.17	0.37	1.27	0.53
	DCTN	740.49	1330.59	1355.33	39.08	39.09	39.10
	MSSCFormer	54.59	181.27	81.59	0.46	1.22	0.63

TABLE XII

AVERAGE TRAINING TIME AND TESTING TIME ON THREE DATASETS

Methods		Training Time (s)			Testing Time (s)		
		HS	Dioni	SA	HS	Dioni	SA
CNN-based methods	FDSSC	0.89	3.76	0.27	3.87	10.85	1.29
	HybridSN	0.40	1.22	3.21	2.86	8.45	26.27
	DBDA	0.63	2.18	0.18	4.42	12.32	1.60
Transformer-based methods	ViT	0.19	1.19	0.24	1.54	8.93	2.11
	SpectralFormer	0.16	0.68	0.20	1.30	5.38	1.79
	MFT	0.15	0.67	0.21	1.09	5.58	1.81
CNN-Transformer-based methods	HybridFormer	0.17	0.53	0.43	1.32	4.01	3.67
	MASSF	0.08	0.61	0.21	0.63	4.45	1.87
	DCTN	0.68	1.26	1.92	4.80	8.57	15.94
	MSSCFormer	0.15	1.04	1.25	1.21	7.71	10.79

methods, Transformer-based methods, and CNN-Transformer-based methods. The details can be found in Section IV-B.

1) Quantitative Comparisons: Tables VIII–X show the quantitative classification performance of the proposed method and the compared methods on the three datasets using the evaluation indicators OA, AA, and k, with the best results in bold. All experimental results are listed in the form of mean values ± standard deviation. Overall, the CNN-based methods perform poorly on the HS and Dioni datasets due

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

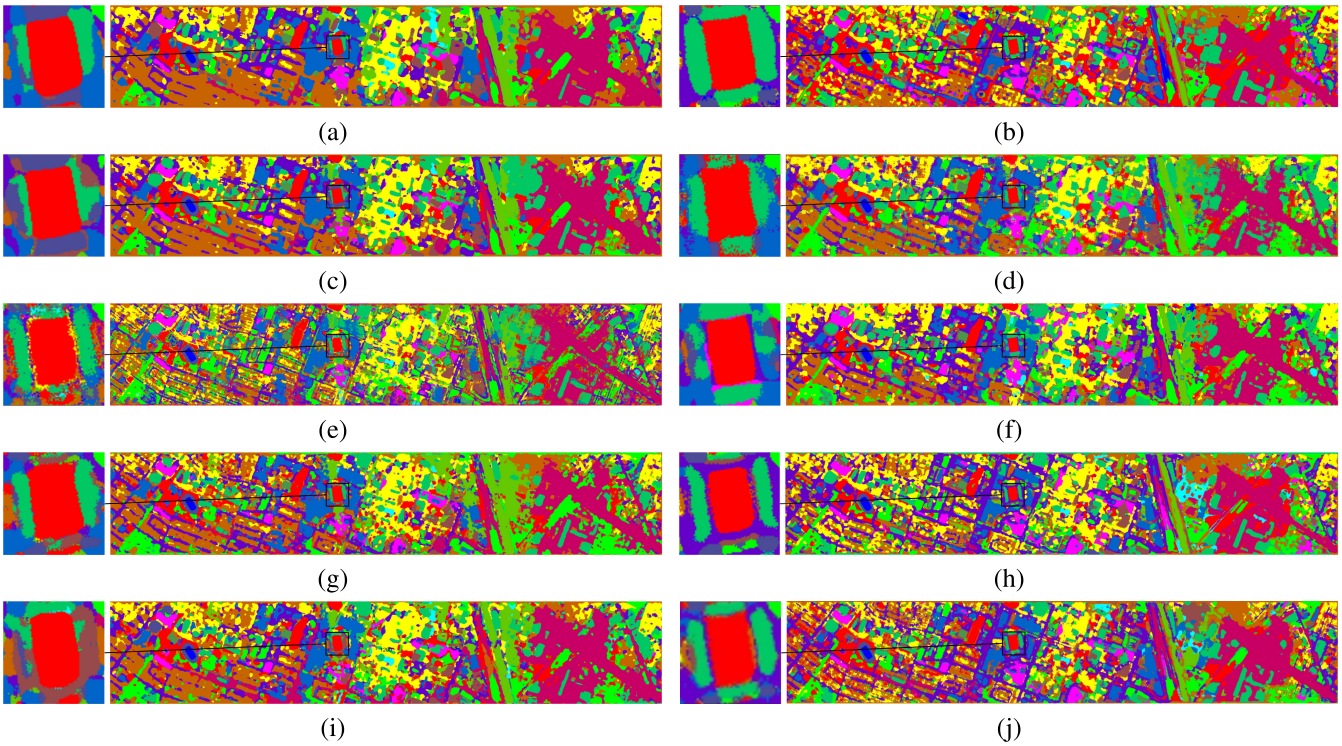


Fig. 6. Classification maps of the Houston2013 dataset. (a) FDSSC, (b) HybridSN, (c) DBDA, (d) ViT, (e) SpectralFormer, (f) MFT, (g) HybridFormer, (h) MASSFormer, (i) DCTN, and (j) MSSCFormer (ours).

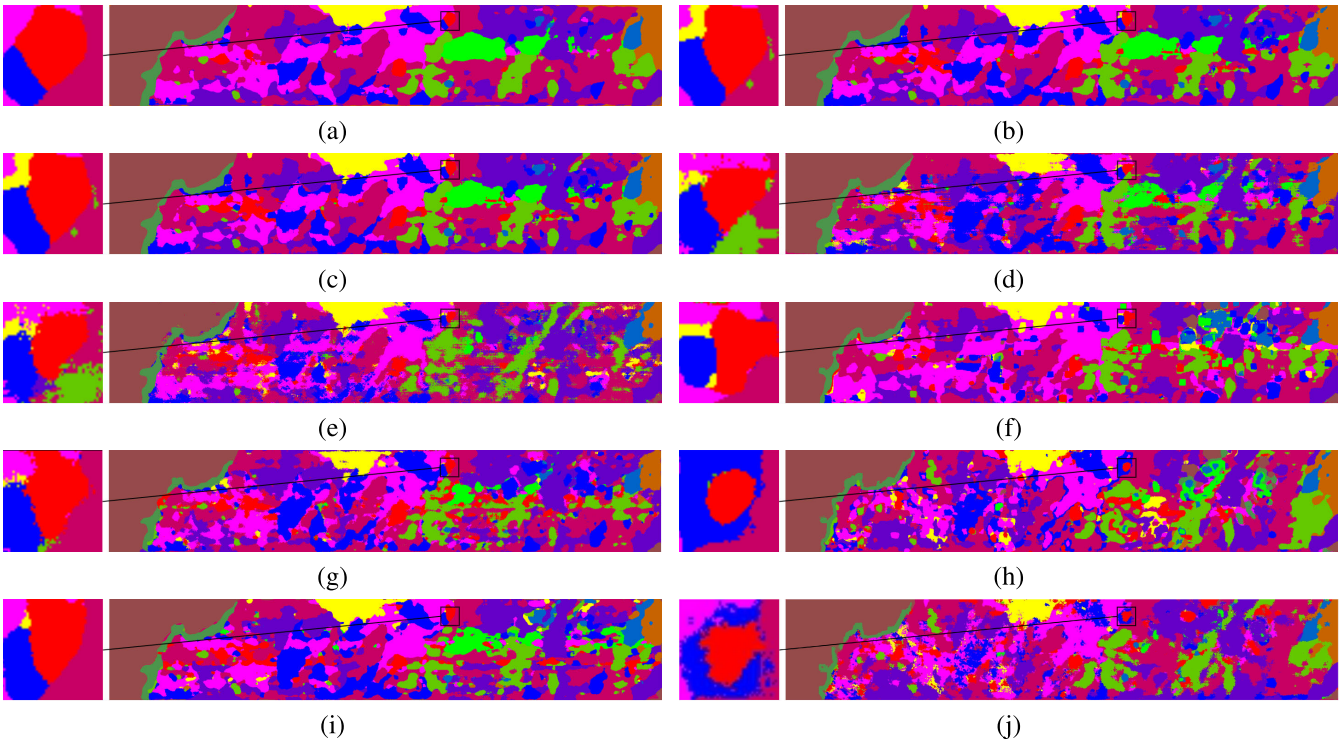
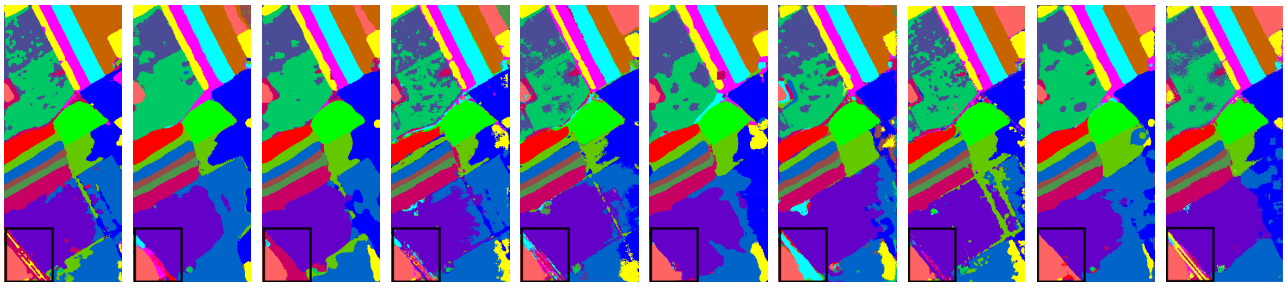


Fig. 7. Classification maps of the Dioni dataset. (a) FDSSC, (b) HybridSN, (c) DBDA, (d) ViT, (e) SpectralFormer, (f) MFT, (g) HybridFormer, (h) MASSFormer, (i) DCTN, and (j) MSSCFormer (ours).

to their lack of global modeling capabilities. Although the Transformer-based methods have the ability to model global image information, they are not effective in processing local detail information in the image, resulting in poor overall performance of this method. The CNN-Transformer based method combines the advantages of the two methods and has the best overall performance on the three datasets. In addition, MSSCFormer achieved the best classification results on all datasets.

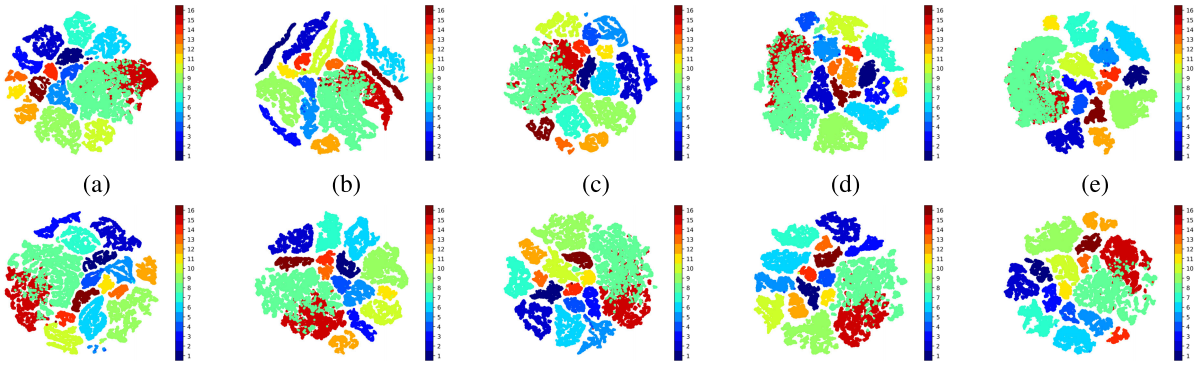
Table VIII lists the classification results on the HS dataset. Only a small number of single-class accuracy optima appear in the CNN-based and Transformer-based methods. Most of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Classification maps of the Salinas dataset. (a) FDSSC, (b) HybridSN, (c) DBDA, (d) ViT, (e) SpectralFormer, (f) MFT, (g) HybridFormer, (h) MASSFormer, (i) DCTN, and (j) MSSCFormer (ours).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. T-SNE maps of different methods on the Salinas dataset. (a) FDSSC (OA = 93.14%), (b) HyBridSN (OA = 93.47%), (c) DBDA (OA = 91.91%), (d) ViT (OA = 89.96%), (e) SpectralFormer (OA = 89.46%), (f) MFT (OA = 91.74%), (g) HyBridFormer (OA = 91.21%), (h) MASSFormer (OA = 93.90%), (i) DCTN (OA = 93.43%), and (j) MSSCFormer (Ours, OA = 94.35%).

single-class optimal values appear in the CNN-Transformer-based methods. The MSSCFormer contains more than half of the optimal values of single-class accuracy and achieves the best classification results in OA, AA, and k metrics, which further validates the superiority of our proposed method.

The classification results of the Dioni dataset are recorded in Table IX, and the accuracies of the three methods are relatively close. However, our proposed method significantly improves the classification performance and achieves the best performance in several single-class accuracies.

As can be seen from Table X, the CNN-based method achieves the best results in multiple single-class accuracies, mainly due to the presence of some small-sample classes in SA. The Transformer-based method has a poor classification performance due to the lack of local detail mining capability of this type of method. The MSSCFormer outperforms all the models.

Taking the OA evaluation metric as an example, the accuracies of MSSCFormer are 2.31%, 4.44%, and 0.45% higher than the best model in the comparison models on the HS, Dioni, and SA datasets, respectively. Overall, MSSCFormer outperforms the other compared models on all three datasets for all three evaluation metrics, proving the superiority of the algorithm.

2) Qualitative Comparisons: To show more intuitively the classification results of the proposed model and other comparative methods on the three datasets, we qualitatively show the classification graphs corresponding to the ten algorithms. Figs. 6–8 show the classification result plots of HS, Dioni, and SA, from which it can be seen that MSSCFormer has the best classification effect. The classification of the three methods is observed by comparing the classification graph with the ground-truth graph corresponding to the image. The CNN-based method produces more misclassifications of the boundary pixels of adjacent land classes. This may be because the characteristics of edge pixels are not obvious, and the local receptive field of the CNN-based method makes it easy to misclassify the original details. Transformer-based methods are more likely to misclassify pixels within the class, which is due to the fact that these methods lack the ability to mine the local details of the image. From the three classification maps, it can be seen that MSSCFormer classifies the pixels more reasonably and produces finer classification maps.

3) Visualization: To further visually compare the performance of MSSCFormer and the comparison models, we conducted a t-SNE visualization comparative analysis of features on the Salinas dataset. Fig. 9 shows the 2-D graphical plots, where different colored dots represent different categories of samples. MSSCFormer demonstrates the best clustering results, which has a tighter distribution in intra-classes and more distinct boundaries in inter-classes. For example, MSSCFormer has the tightest clustering result in the two largest and challenging classes, classes 8 and 15. One possible reason for this is that ICSA achieves smaller differences in the intra-class samples and larger differences in the inter-class samples. On the other hand, the local-global-local model structure extracts more recognizable local and global features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

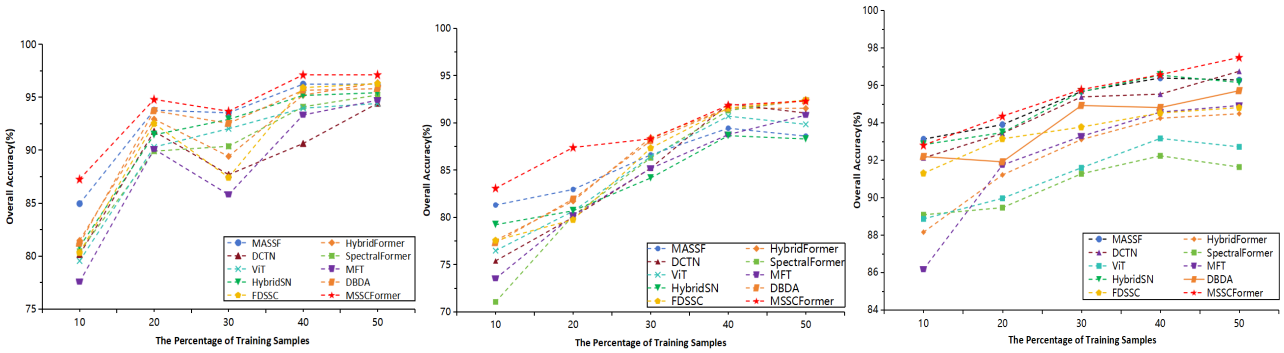


Fig. 10. Overall classification accuracy of various algorithms under different training sample sizes. (a) Houston2013. (b) Dioni. (c) Salinas.

4) Complexity and Efficiency Analysis: To further analyze the effectiveness of MSSCFormer, we evaluated the model complexity of ten algorithms on three datasets, including the number of parameters (Param), floating-point operations (FLOPs), training time, and test time. We recorded the average training time of 50 training epochs as our training time and average inference time from 20 repeated experiments. As shown in Table XI, MSSCFormer has advantages over most models in all methods in terms of both parameter quantity and FLOPs. This can be attributed to a small number of encoders and DWConv in SSFE and GLCT, which reduces a large number of parameters while maintaining the modeling capability of spatial–spectral features. Meanwhile, as shown in Table XII, the SpectralFormer, MFT, and MASSF exhibited shorter training and testing times than the remaining models, which can be attributed to their relatively simple architecture, lacking hierarchical structures. One reason for the slow inference of MSSCFormer is that the parallel convolutional structure of LCFB adds intermediate feature caching and multicore concurrent scheduling burdens. Another is the global attention operation of MHSA in the inference phase. Although MSSCFormer has no obvious advantages in training speed and test speed, it achieves the best classification performance on all datasets.

F. Influence of Training Samples

To verify the scalability of MSSCFormer, we experimented on three datasets to show how different numbers of training samples affect model performance. We selected a set of training samples with a number of 10, 20, 30, 40, and 50, and the remaining samples were used as test samples. OA was used as the evaluation metric for the classification results.

As shown in Fig. 10, MSSCFormer still demonstrates superior classification accuracy compared to other models with increasing training samples. Specifically, as the training samples increase, the classification accuracy of the MSSCFormer method gradually increases on all three datasets, except for the case where the training samples are 30 on the HS dataset. From Fig. 10(a), it can be seen that several models have the problem of accuracy degradation in the case of HS sample size of 30. The main reason may be that the randomness of the dataset division leads to the low quality of the selected samples. This indicates that the method has relatively great generalization ability and stability. The classification accuracy of MSSCAFormer is lower than that of some models only when the training samples are 40 and 50 on the Dioni dataset, but in the vast majority of cases, the classification accuracies are more than that of all the selected comparison models. It can be seen that our method demonstrates strong classification performance in the case of limited training samples, which is also a major feature of the model. Overall, the proposed model has excellent performance and good scalability.

V. Conclusion

In this article, we propose a novel local-global-local hybrid CNN-Transformer-based model, MSSCFormer, for the HSI classification task. To better identify the same class of land samples, the method introduces ICSA to reduce the spectral differences between samples of the same land class, which also provides a great basis for the model to learn more discriminative features. In addition, we designed SSFE and GLCT. SSFE consists of a 3-D DSConv with residual concatenation and MGSFF. The former is used to extract shallow spatial–spectral features of HSI, while the latter is used to extract multigranularity spectral features. GLCT uses the central pixel of the sequence as the classification object and captures the global dependencies between pixels through MHSA from a global perspective. To compensate for the lack of Transformer's ability to mine local information, we introduce the LCFB in GLCT to extract multigranularity local information from a local perspective. Experiments on three public datasets demonstrate the effectiveness and generalization of the proposed method.

In future work, we will explore the application of the lightweight model in scenarios with a limited number of samples. Also, semi-supervised, unsupervised, and transfer learning will be explored for resource-limited scenarios.

References

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[3] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

[4] P. S. Thenkabail, J. G. Lyon, and A. Huete, "Advances in hyperspectral remote sensing of vegetation and agricultural crops," in *Fundamentals, Sensor Systems, Spectral Libraries, and Data Mining for Vegetation*. Boca Raton, FL, USA: CRC Press, 2018, pp. 3–37.

[5] L. Zhang, L. Zhang, D. Tao, X. Huang, and B. Du, "Hyperspectral remote sensing image subpixel target detection based on supervised metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4955–4965, Aug. 2014.

[6] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[7] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[8] G. Hughes, "On the mean accuracy of statistical pattern recognizers," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 1, pp. 55–63, Jan. 1968.

[9] G. M. Foody, "Status of land cover classification accuracy assessment," *Remote Sens. Environ.*, vol. 80, no. 1, pp. 185–201, Apr. 2002.

[10] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.

[11] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.

[12] J. Jiang, J. Ma, C. Chen, Z. Wang, Z. Cai, and L. Wang, "SuperPCA: A superpixelwise PCA approach for unsupervised feature extraction of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4581–4593, Aug. 2018.

[13] J. Wang and C.-I. Chang, "Independent component analysis-based dimensionality reduction with applications in hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1586–1600, Jun. 2006.

[14] S. Jia et al., "Flexible Gabor-based superpixel-level unsupervised LDA for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10394–10409, Dec. 2021.

[15] Y. Gu, Y. Liu, and Y. Zhang, "A selective KPCA algorithm based on high-order statistics for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 1, pp. 43–47, Jan. 2008.

[16] W. Sun, G. Yang, B. Du, L. Zhang, and L. Zhang, "A sparse and low-rank near-isometric linear embedding method for feature extraction in hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 4032–4046, Jul. 2017.

[17] Y. Duan, H. Huang, and Y. Tang, "Local constraint-based sparse manifold hypergraph learning for dimensionality reduction of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 613–628, Jan. 2021.

[18] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[19] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[20] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.

[21] C. Cariou and K. Chehdi, "A new k-nearest neighbor density-based clustering method and its application to hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 6161–6164.

[22] L. Mou, P. Ghamisi, and X. X. Zhu, "Unsupervised spectral–spatial feature learning via deep residual conv–deconv network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 391–406, Jan. 2018.

[23] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[24] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[25] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[26] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.

[27] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Hyperspectral image classification using spectral–spatial LSTMs," *Neurocomputing*, vol. 328, pp. 39–47, Feb. 2019.

[28] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[29] Y. Ding et al., "Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536716.

[30] Y. Ding et al., "Adaptive homophily clustering: Structure homophily graph learning with adaptive filter for hyperspectral image," 2025, *arXiv:2501.01595*.

[31] Y. Ding et al., "SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images," 2025, *arXiv:2502.03497*.

[32] J. Feng et al., "S^4DL: Shift-sensitive spatial–spectral disentangling learning for hyperspectral image unsupervised domain adaptation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 16, 2025, doi: 10.1109/TNNLS.2025.3556386.

[33] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, pp. 1–12, 2015.

[34] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[35] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[36] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Mar. 31, 2025, doi: 10.1109/TCSVT.2025.3556548.

[37] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[38] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[39] L. Yang et al., "FusionNet: A convolution–transformer fusion network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 16, p. 4066, Aug. 2022.

[40] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[41] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[42] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[43] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[44] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[45] Y. Zhang, L. Liang, J. Mao, Y. Wang, and L. Jia, "From global to local: A dual-branch structural feature extraction method for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1778–1791, 2025.

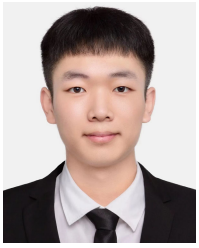
[46] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[47] Y. Qing, W. Liu, L. Feng, and W. Gao, "Improved transformer net for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 11, p. 2216, Jun. 2021.

[48] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.

[49] Z. Xue, Q. Xu, and M. Zhang, "Local transformer with spatial partition restore for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4307–4325, 2022.
[50] Y. Xu et al., "Spatial–spectral 1DSwin transformer with groupwise feature tokenization for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5516616.
[51] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-mamba: Interval group spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.
[52] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.
[53] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.
[54] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.
[55] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.
[56] L. Liang, Y. Zhang, S. Zhang, J. Li, A. Plaza, and X. Kang, "Fast hyperspectral image classification combining transformers and SimAM-based CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522219.
[57] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.
[58] L. Sun, H. Zhang, Y. Zheng, Z. Wu, Z. Ye, and H. Zhao, "MASSFormer: Memory-augmented spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516415.
[59] N. Chen, L. Fang, Y. Xia, S. Xia, H. Liu, and J. Yue, "Spectral query spatial: Revisiting the role of center pixel in transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402714.
[60] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.
[61] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[62] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.



Gang Chen received the B.S. degree in computer science and technology from Hefei Normal University, Hefei, China, in 2023. He is currently pursuing the M.S. degree in artificial intelligence with Guangxi University, Nanning, China.

His research interests include deep learning, hyperspectral image classification, optical character recognition, and object detection.



Lina Yang (Member, IEEE) received the B.S. degree in computer engineering from Shijiazhuang Railway University, Shijiazhuang, China, in 2005, the M.S. degree in computer science from the University of Malaya, Kuala Lumpur, Malaysia, in 2011, and the Ph.D. degree from the University of Macau, Macau, in 2015.

She is currently an Associate Professor with the School of Computer, Electronics, and Information, Guangxi University, Nanning, China. Her research interests include medical biometrics, machine learning, image processing, and information security.



Thomas Wu (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 1999.

He then joined as an Assistant Professor with the University of Central Florida, Orlando, FL, USA, in 1999, and was promoted to an Associate Professor in 2005 and a Professor in 2011. He also got his tenure in 2005. He is currently the Dean of the School of Electrical Engineering, Guangxi University, Nanning, China. His research interests include machine learning, high-performance motors, and intelligent equipment.



Lianhui Liang (Member, IEEE) received the M.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2019 and 2024, respectively.

From 2022 to 2024, he was a Visiting Ph.D. Student at the Hyperspectral Computing Laboratory, University of Extremadura, Badajoz, Spain, supported by the China Scholarship Council. He is currently an Assistant Professor with the School of Electrical Engineering, Guangxi University, Nanning, China. His research interests include hyperspectral image processing, signal processing, pattern recognition, and machine learning.



Xichun Li received the B.S. degree in computer science from Northeastern University, Shenyang, China, in 1997, and the M.S. and Ph.D. degrees in computer science from the University of Malaya, Kuala Lumpur, Malaysia, in 2002 and 2010, respectively.

He is currently a Professor of Guangxi Normal University for Nationalities, Guangxi, China. His research interest is mobile computing.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.S. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisbon, Lisbon, Portugal, in 2011.

She is currently a Full Professor with China University of Geosciences, Wuhan, China. Her main research interests include remotely sensed hyperspectral image analysis, signal processing, supervised/semisupervised learning, and active learning.

Prof. Li is the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been a Guest Editor of several journals, including the *Proceedings of the IEEE* and the *ISPRS Journal of Photogrammetry and Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:30 UTC from IEEE Xplore. Restrictions apply.